

Building TRANSLAMORE

Lessons Learned from a Side Project

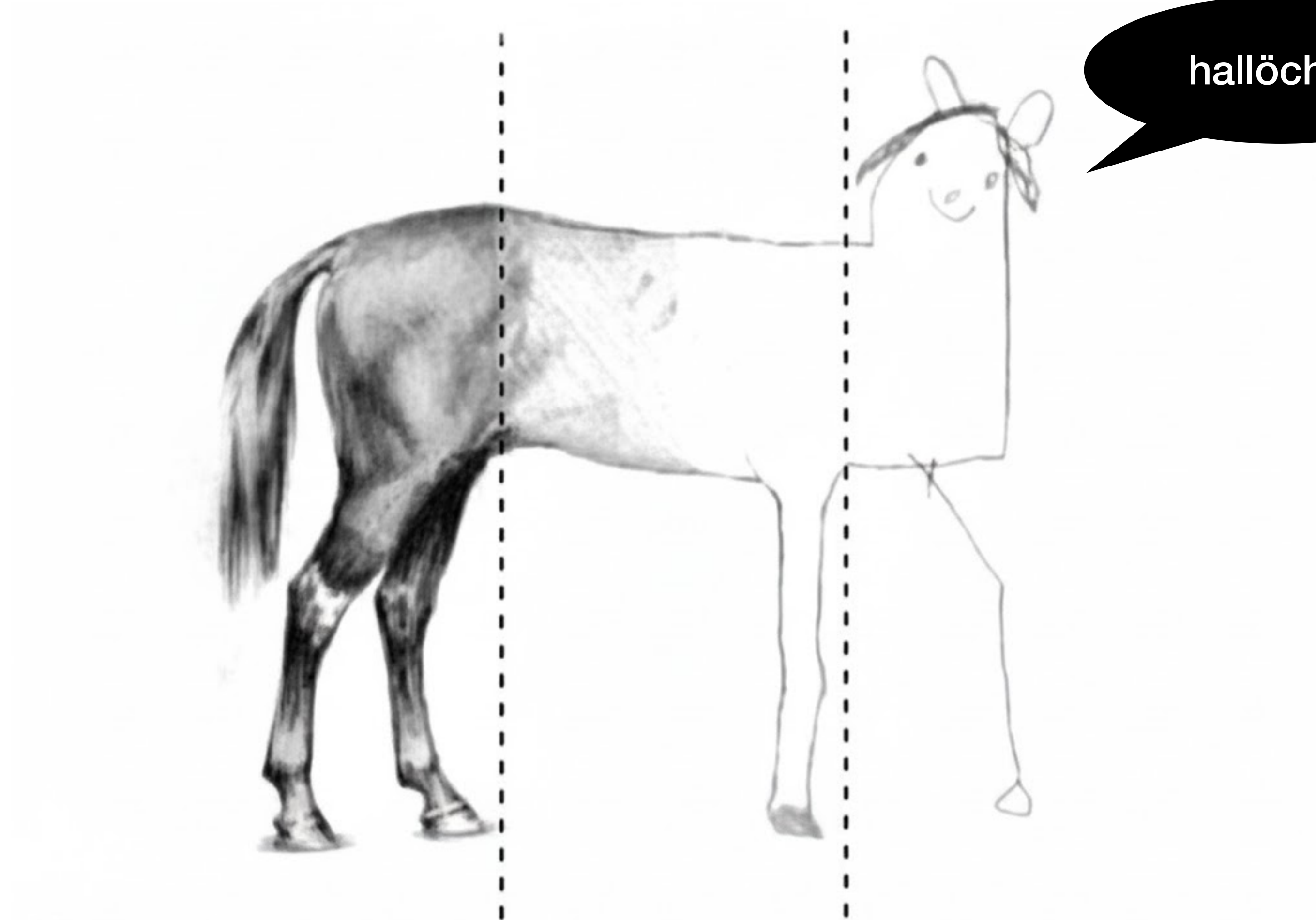
Łukasz Wiśniewski, 11 Dec 2025

#OpenToWork GmbH

Flutter Berlin Meetup x ENPAL



Mein Deutsch



hallöchen

Lesen

Hören

Sprechen



TRANSLAMORE

more than translation...

Reader App

Discover new words

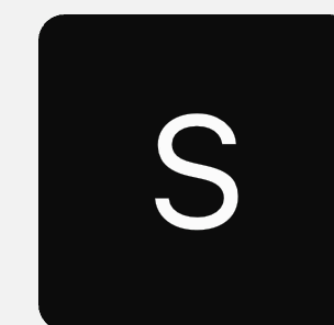
Solaranlagen ermöglichen eine nachhaltige Stromerzeugung direkt vom eigenen Dach. Mit einem Batteriespeicher kann überschüssige Energie effizient

Abhängigkeit *noun*
dependency, dependence,
addiction, reliance



α

Abhängigkeit vom öffentlichen Stromnetz spürbar. Gleichzeitig erhöhen moderne Energiemanagement-



Solarenergie und Speicherl...

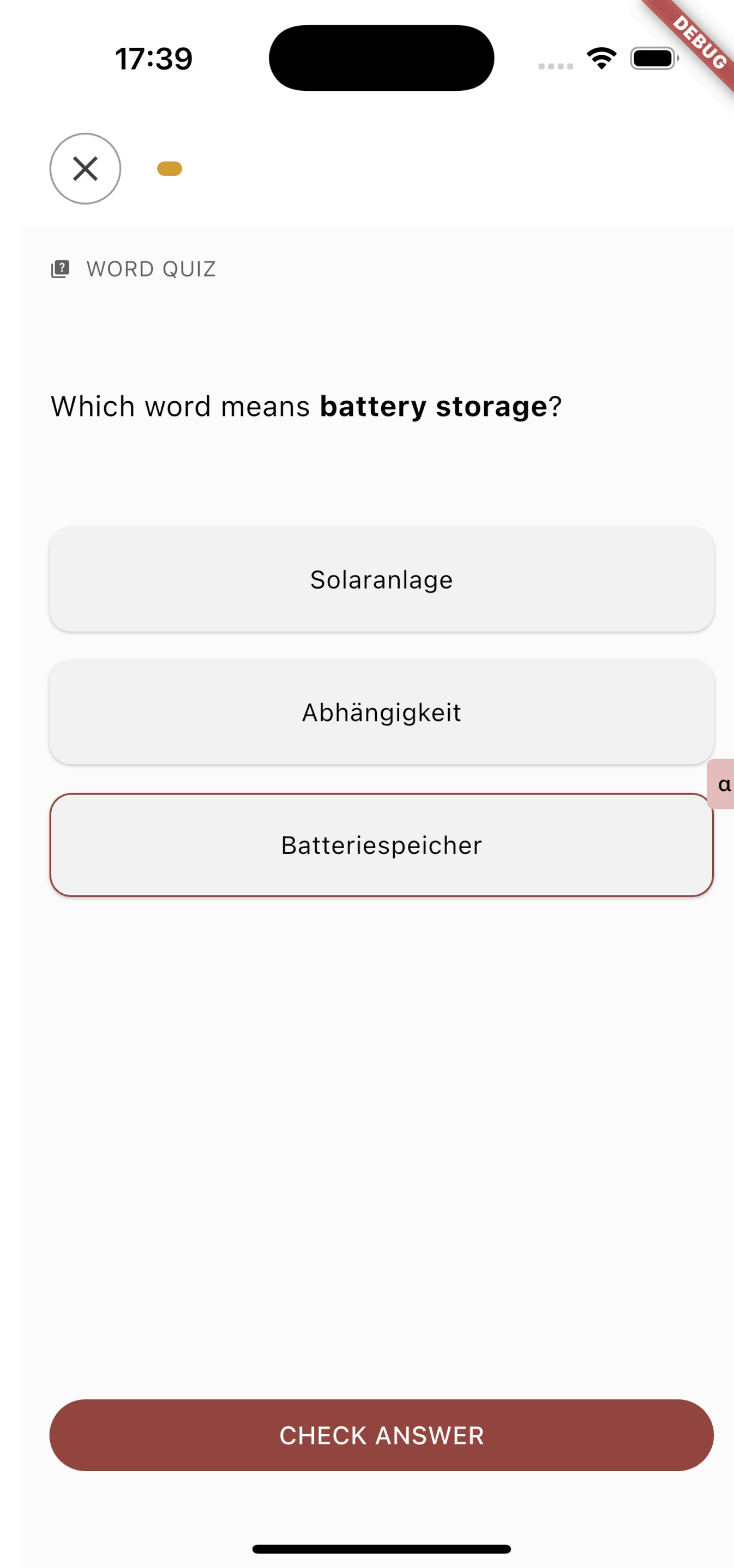


READ LATER

gesamten
Haushaltsverbrauchs

...& Practice App

Contextual Exercises



Language Pairings

English not required

17:19

DEBUG

What's your **native language**?

Which language do you want to **learn**?

Czech

English

Czech

French

French

Use learning language as the app's language.

Continue

Problems with a side project

...which is using AI

- Project management
- Is this thing even useful?
- Boring repetitive tasks
- LLaMent in the cloud
 - Predictability
 - Independence
 - Keeping secrets secret

Project Management

~~Project~~ Management

Motivation Management

Motivation



Time

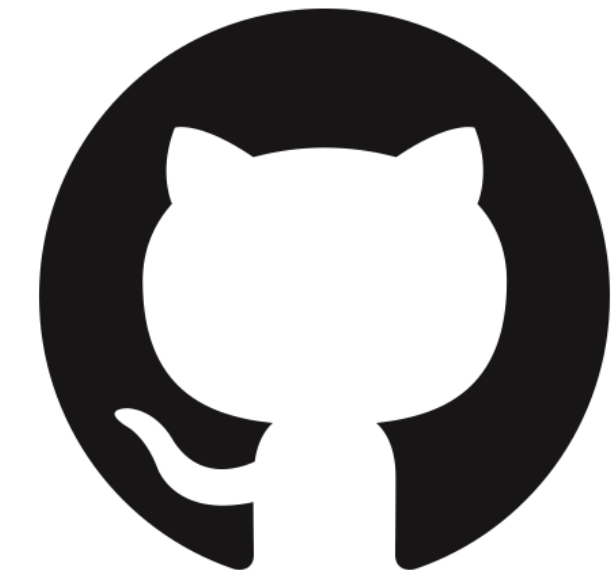




GitLab



Linear



Jira



Trello





Task

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Bug

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Done

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VocabularyWordScreen



Allow removal of all used forms

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Display sentences where used forms are used

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Conjugation tenses are sometimes missing

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Vocabulary Editor

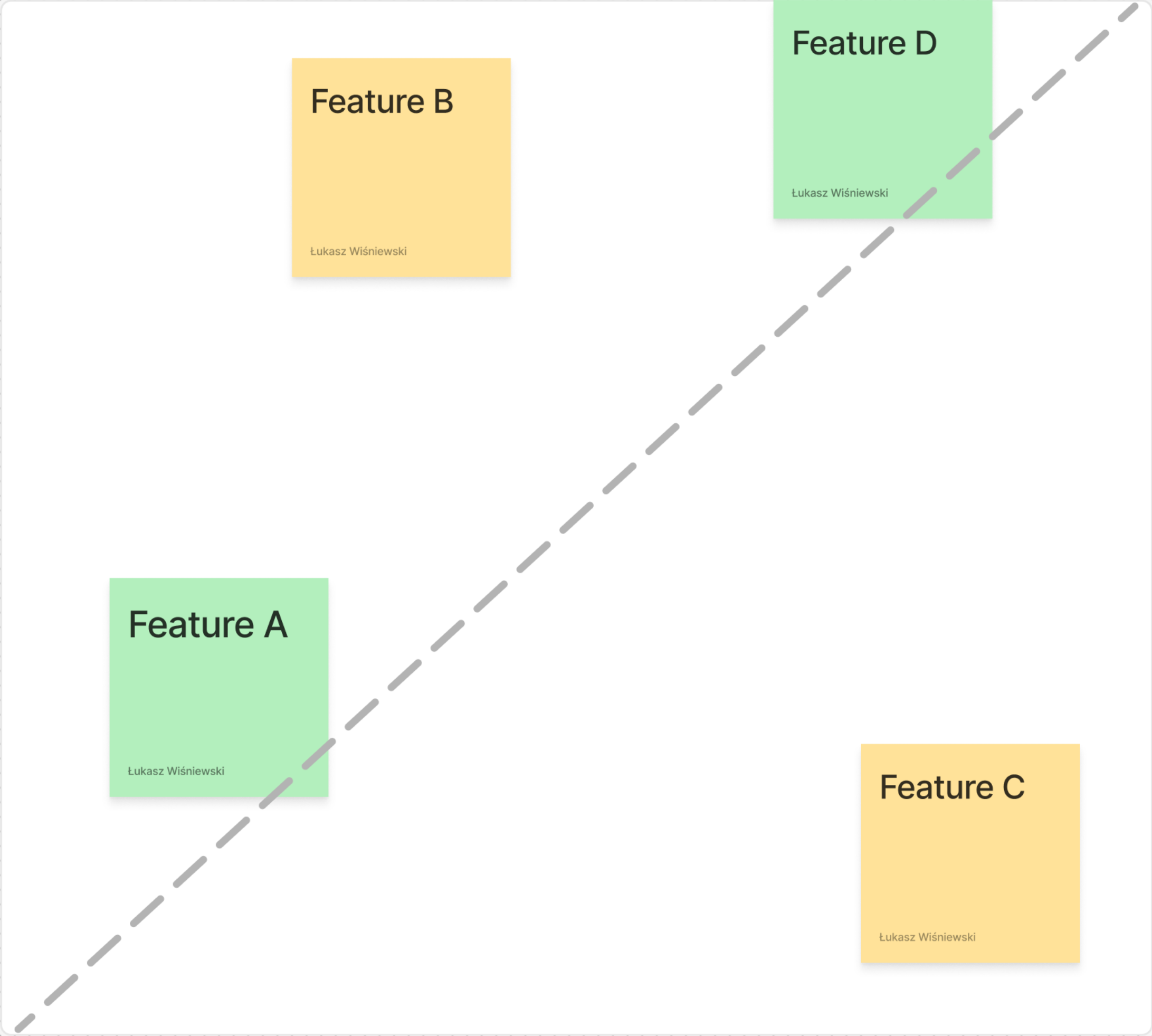
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Breakdown into section based on markdown paragraphs

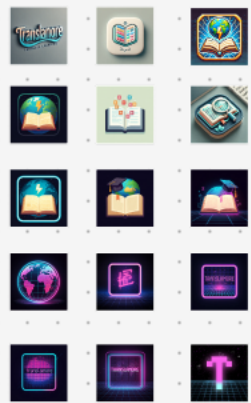
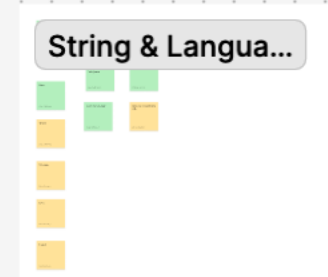
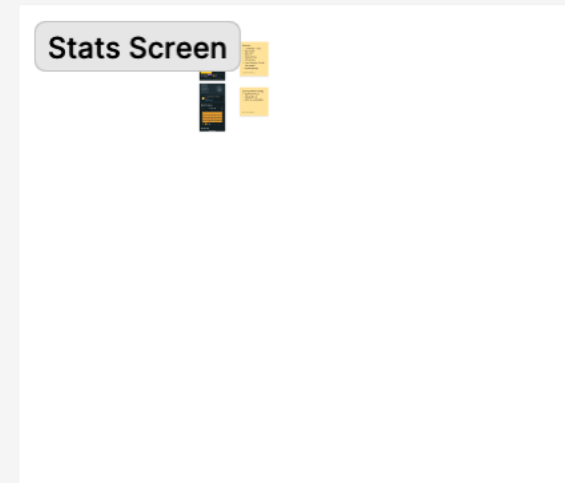
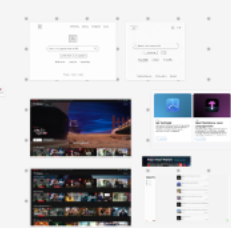
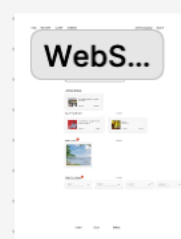
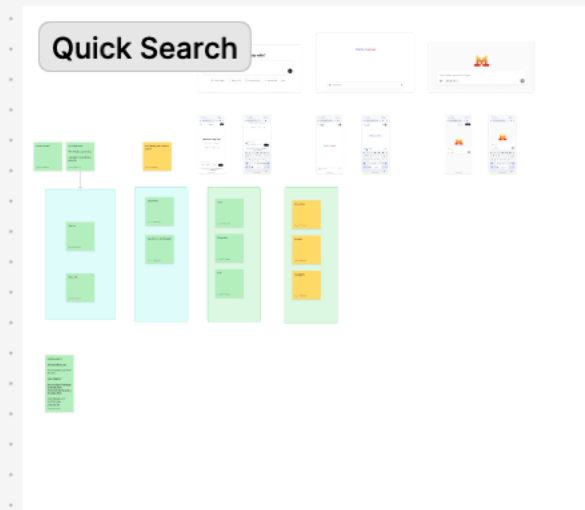
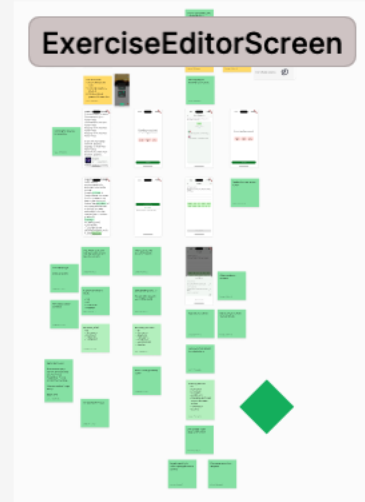
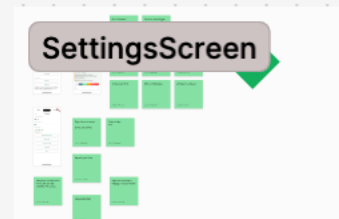
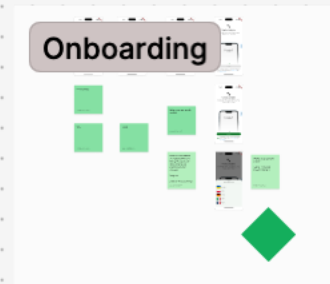
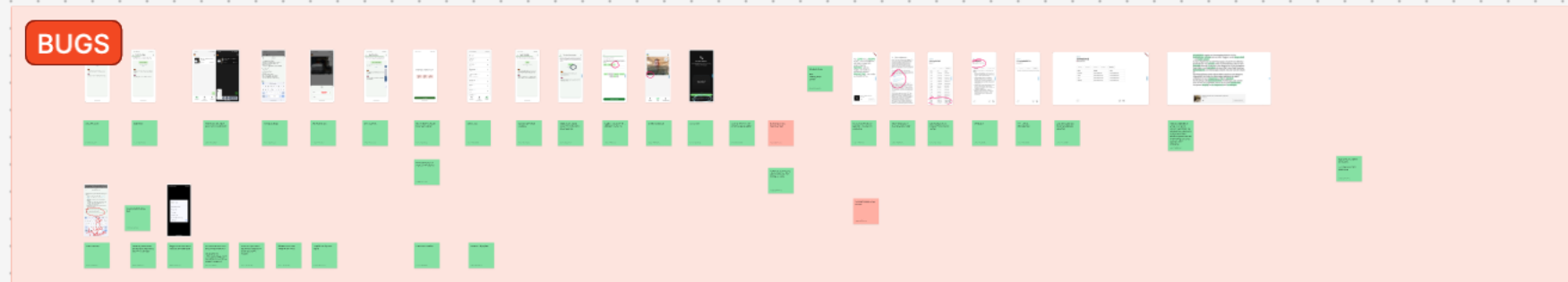
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IMPACT

Effort vs Impact



EFFORT

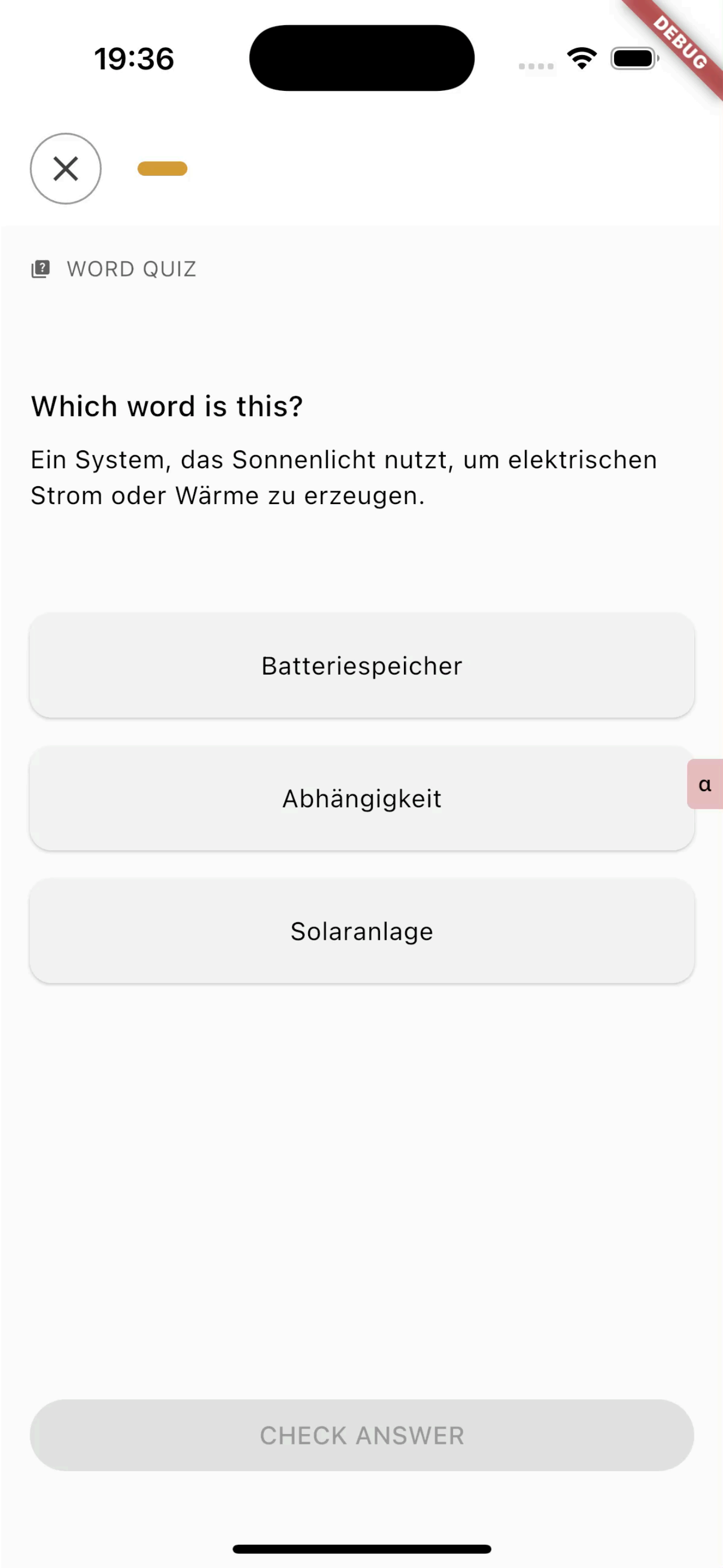
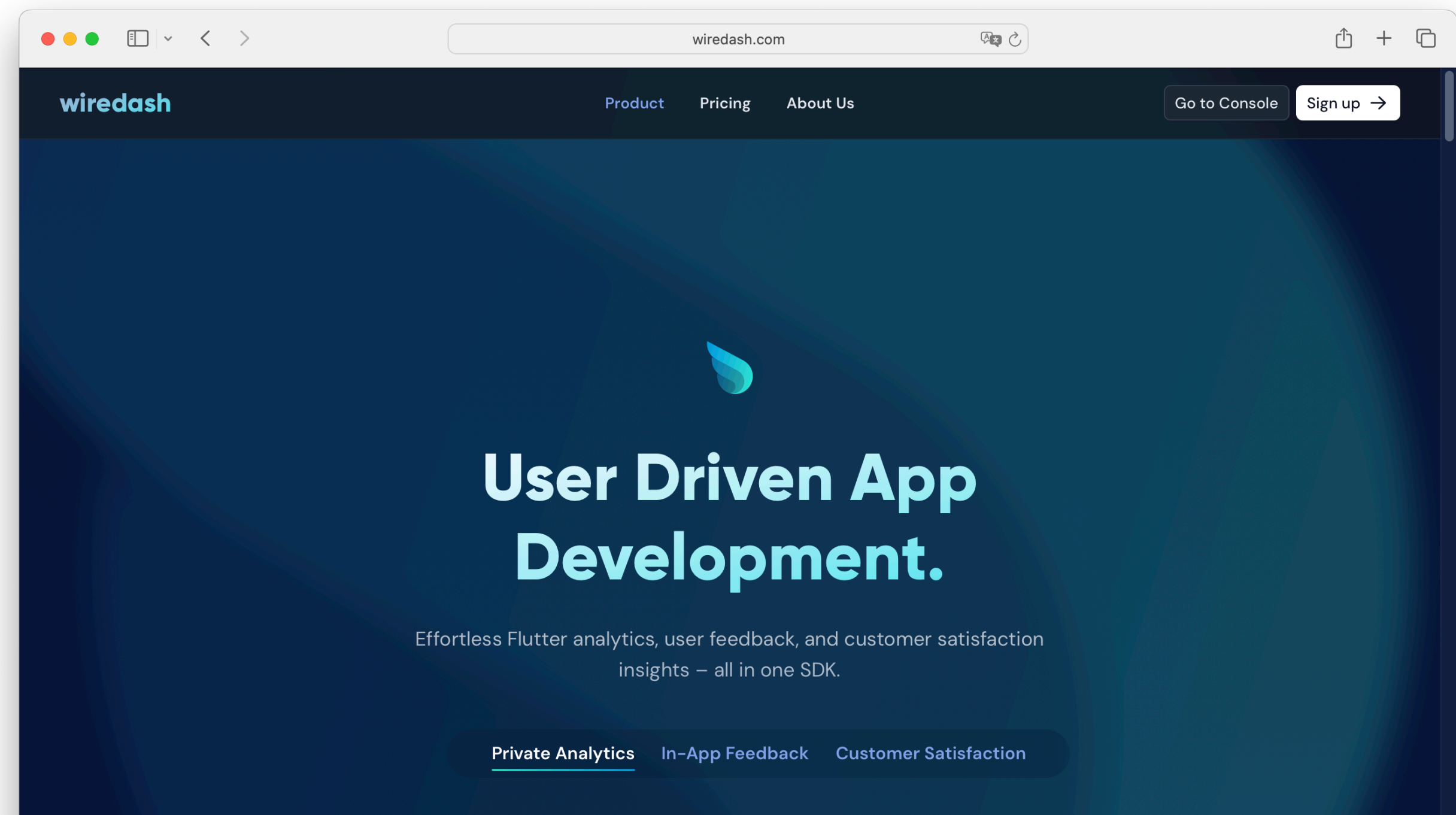


UI/UX is hard



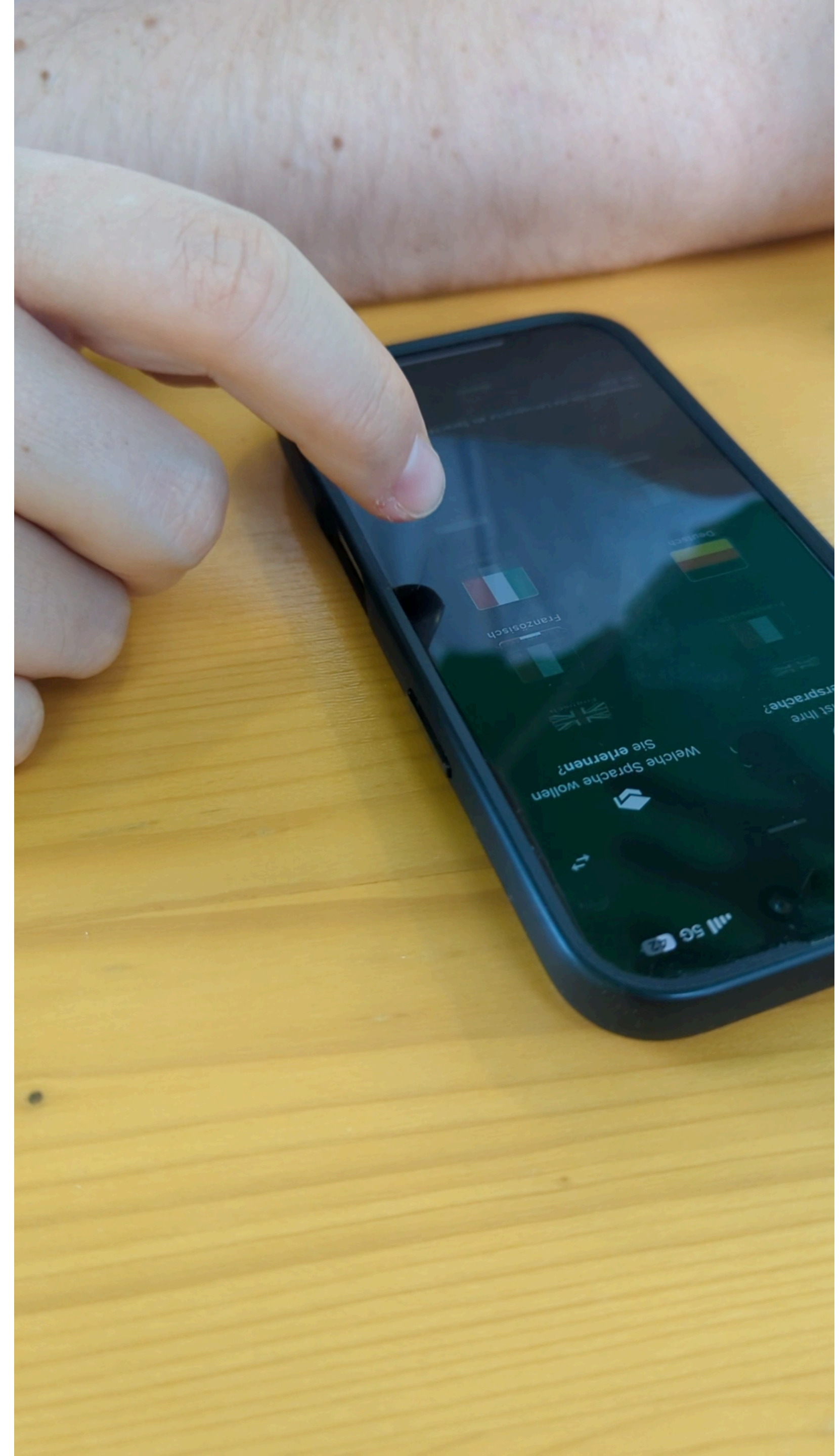
Feedback

- Wiredash



Feedback

- User (Friend) Testing
 - “Oh I didn’t know you can add your own content”
 - “Needs some entry content”
 - “Great product otherwise!”



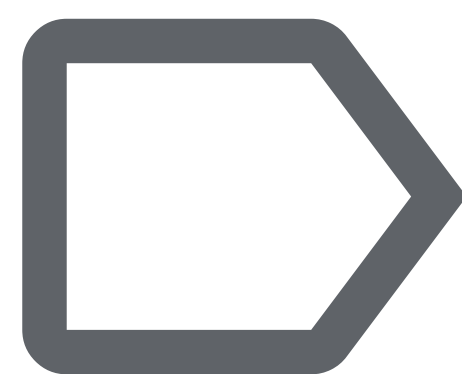
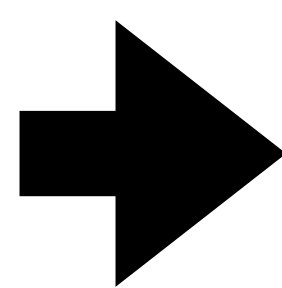
Flutter works everywhere
...but distribution is a PITA

Release Action

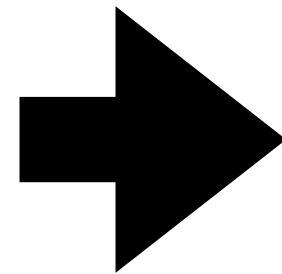


Release Action

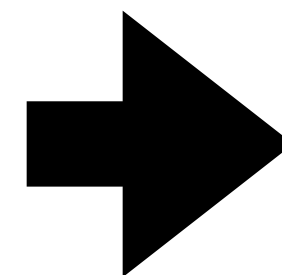
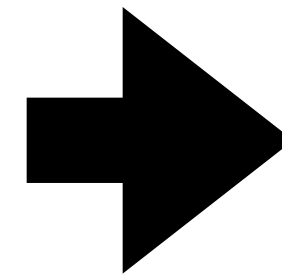
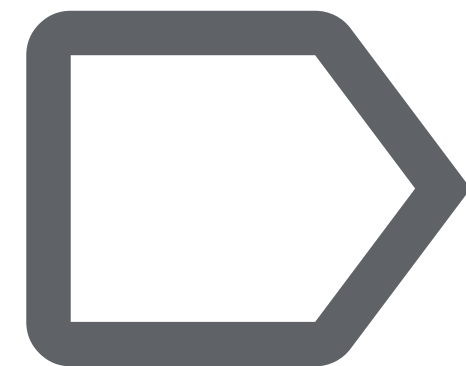
v0.1.121



Release Action



v0.1.121



Digital Ocean



Firebase Hosting



Codemagic

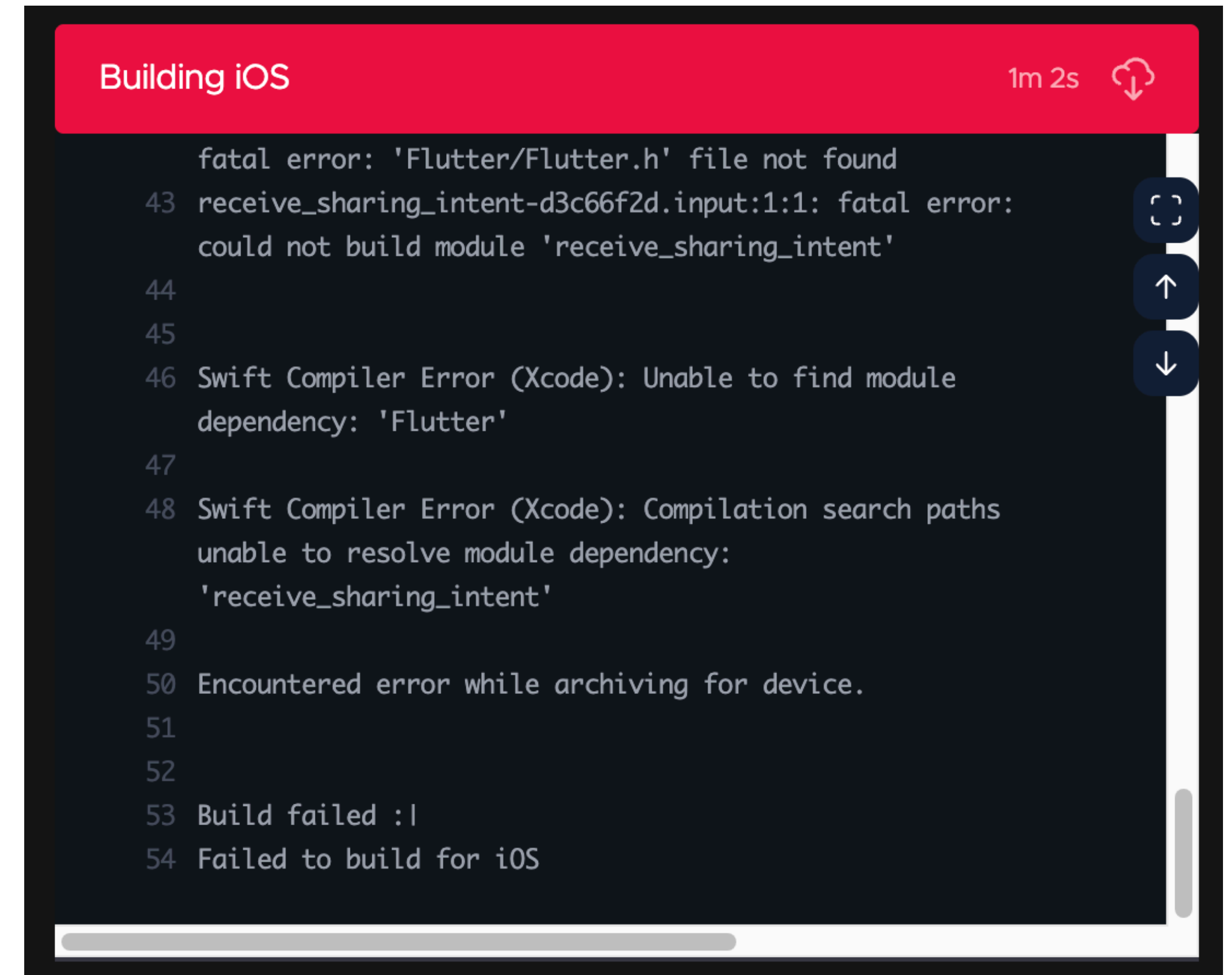
iOS



CI/CD

Continuous surprises every now and then

- Web/Server 🥰
- Google Play 🙄
- App Store 😐
 - TestFlights can expire
- ...still worth investing your time

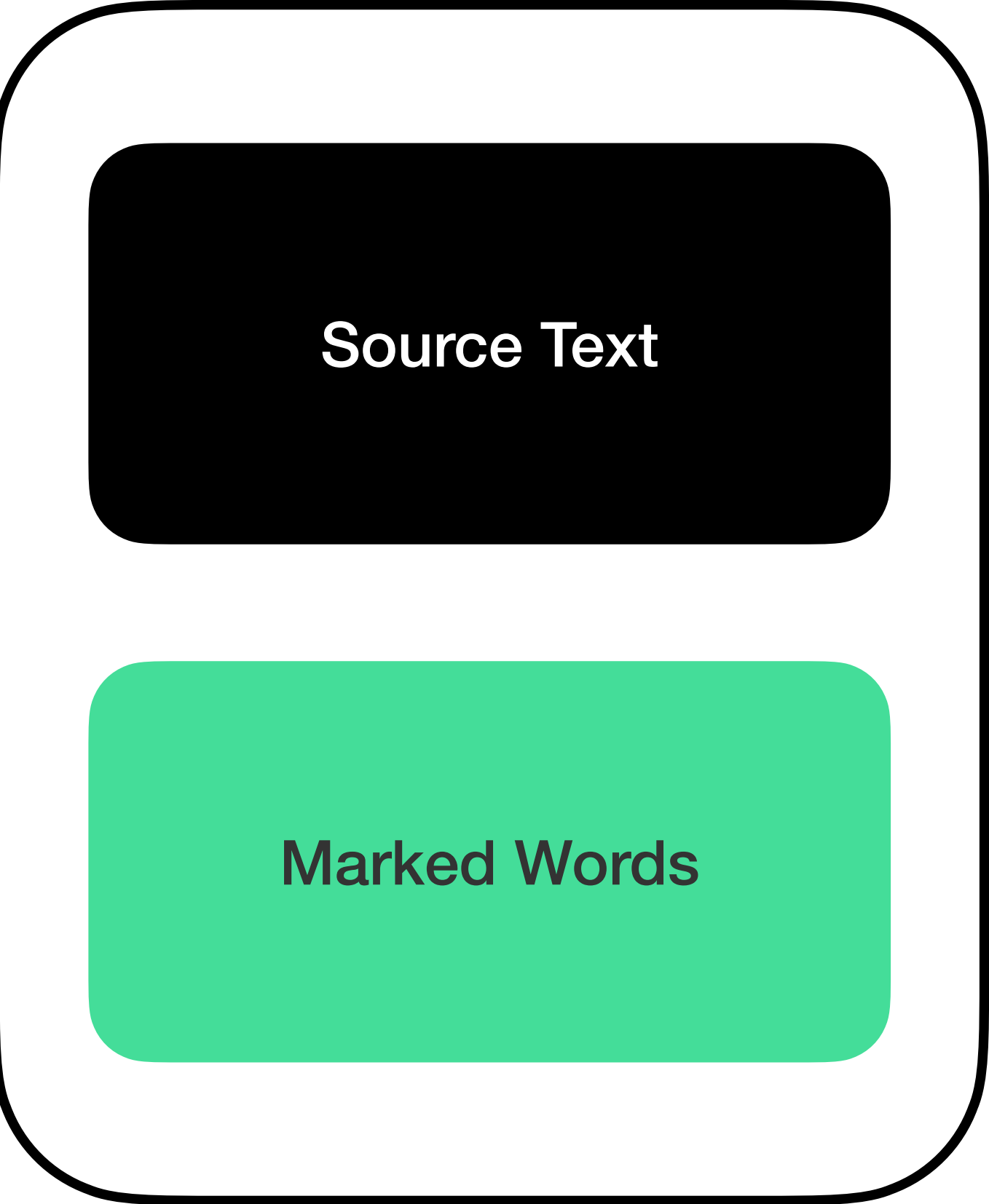


The screenshot shows a terminal window titled "Building iOS" with a red header bar. The terminal output displays a fatal error: "fatal error: 'Flutter/Flutter.h' file not found". This is followed by a line indicating a fatal error in the "receive_sharing_intent" module. The error is then repeated in Swift Compiler Error (Xcode) format, stating "Unable to find module dependency: 'Flutter'" and "Compilation search paths unable to resolve module dependency: 'receive_sharing_intent'". The build process then encounters an error while archiving for the device, resulting in a failed build for iOS. The terminal window includes a scrollbar on the right and a status bar at the bottom showing "1m 2s" and a refresh icon.

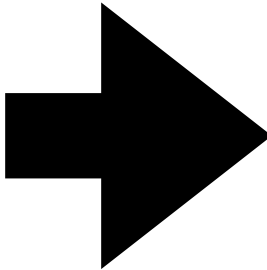
```
Building iOS 1m 2s
fatal error: 'Flutter/Flutter.h' file not found
43 receive_sharing_intent-d3c66f2d.input:1:1: fatal error:
   could not build module 'receive_sharing_intent'
44
45
46 Swift Compiler Error (Xcode): Unable to find module
   dependency: 'Flutter'
47
48 Swift Compiler Error (Xcode): Compilation search paths
   unable to resolve module dependency:
   'receive_sharing_intent'
49
50 Encountered error while archiving for device.
51
52
53 Build failed :|
54 Failed to build for iOS
```

LLM response predictability

input.json



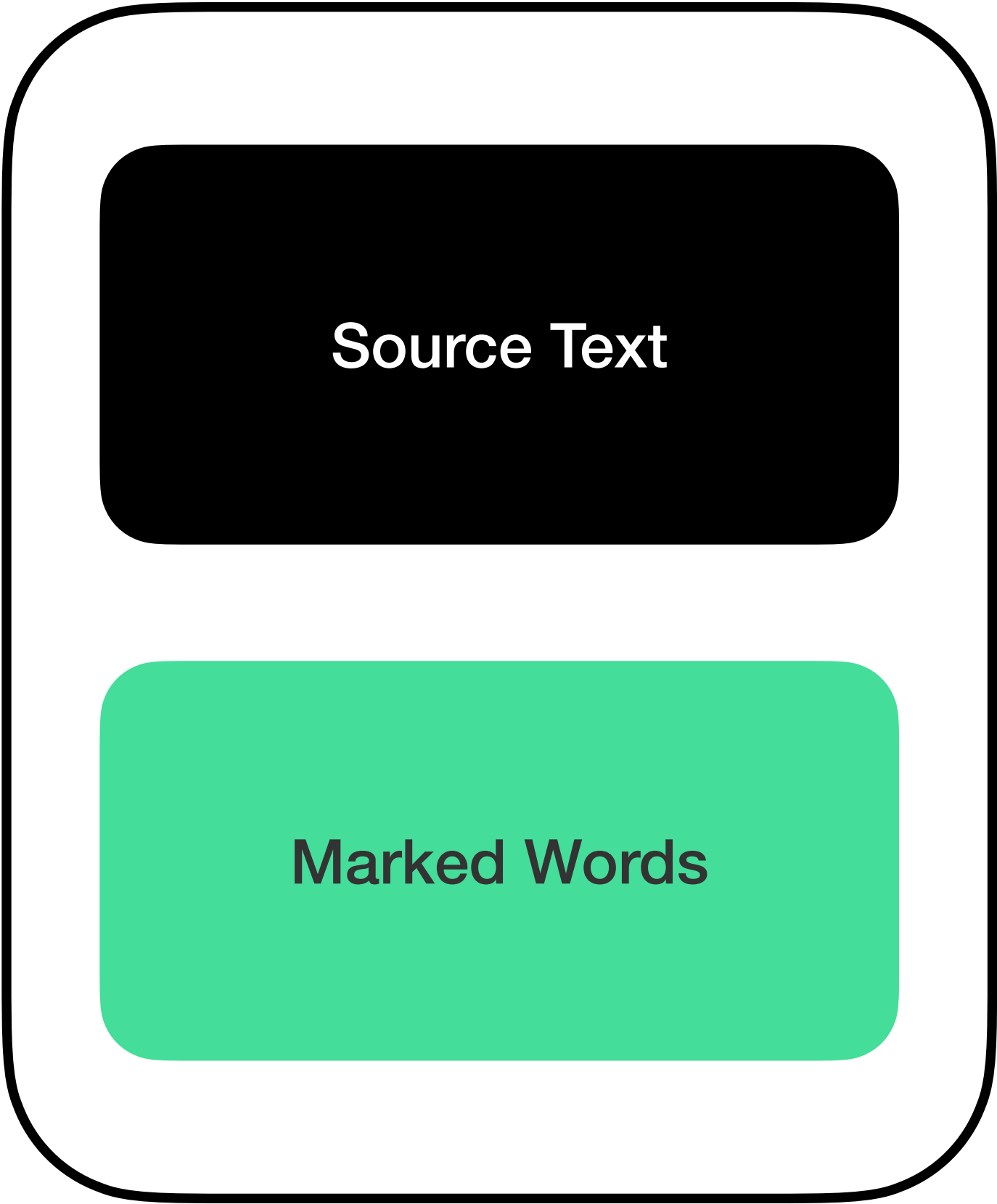
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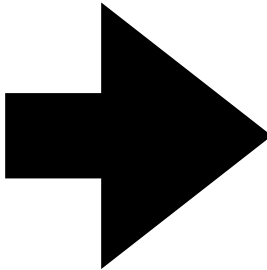
output.json

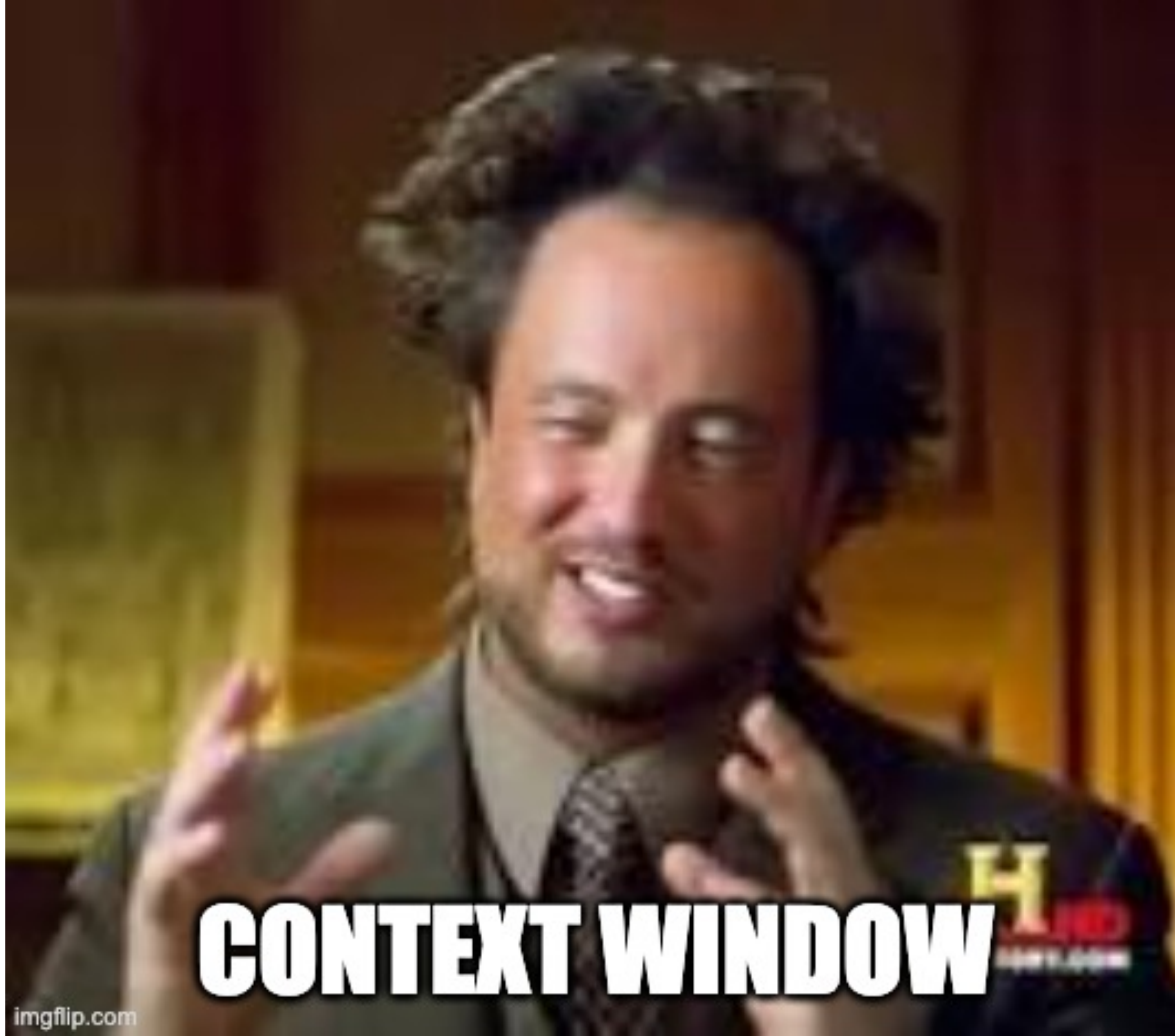


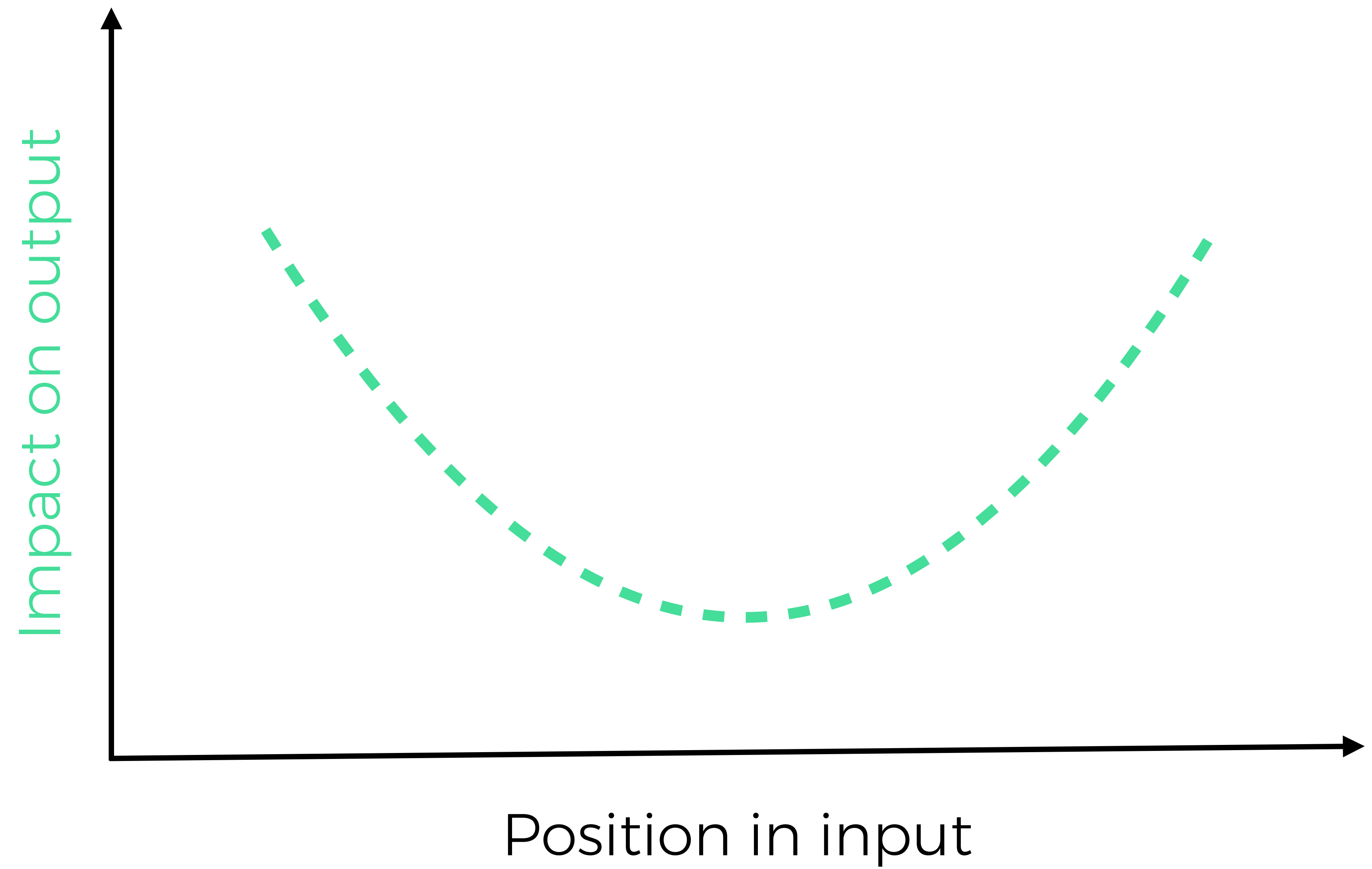
input.json

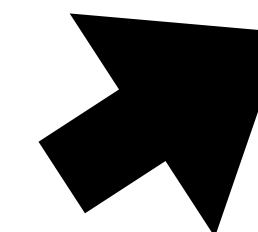
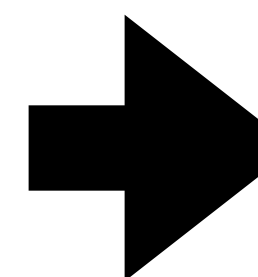
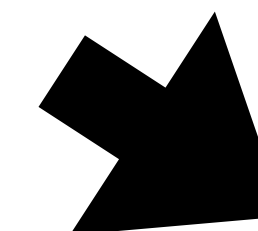
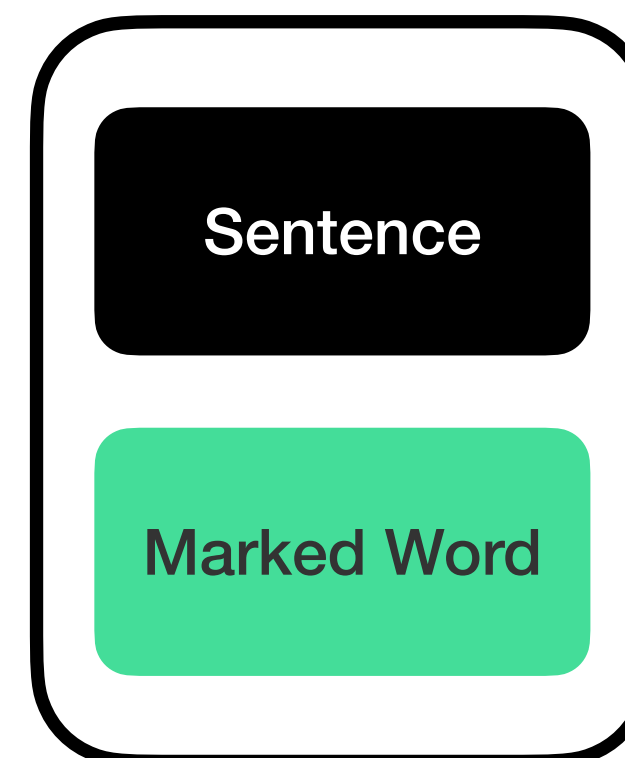
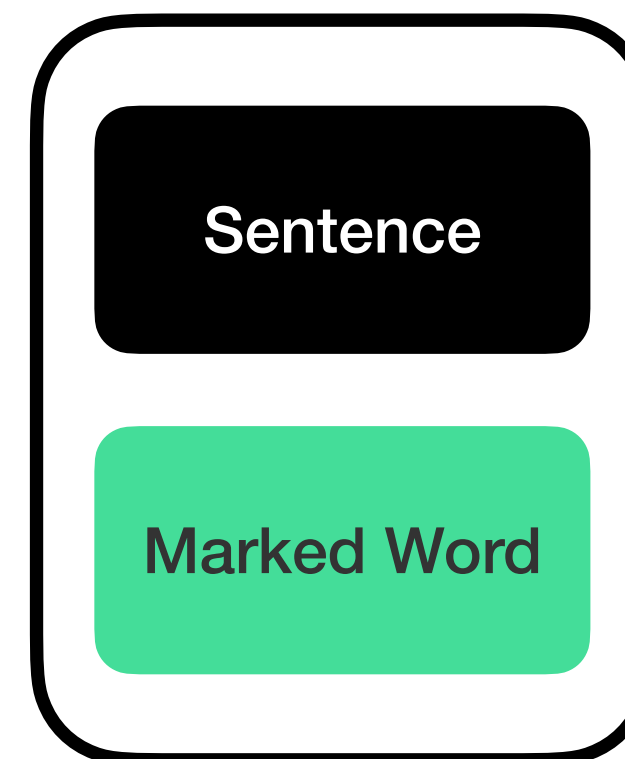
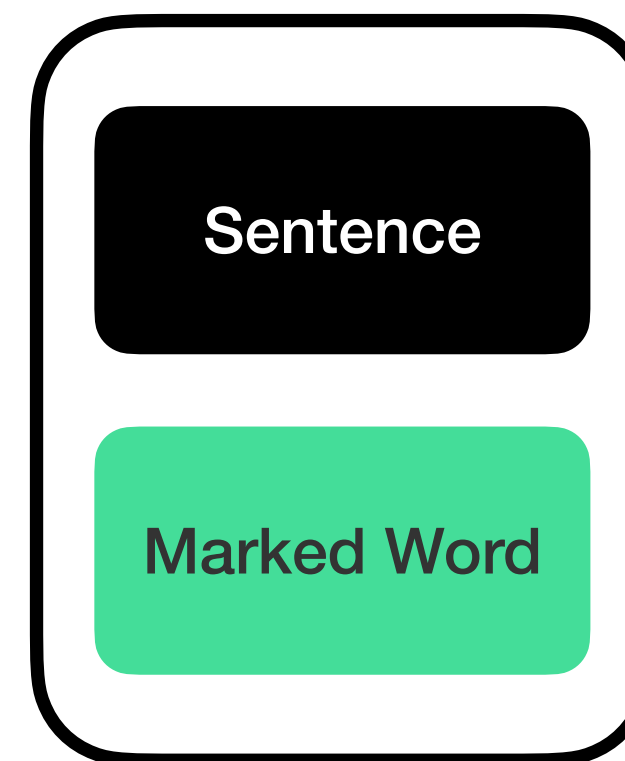
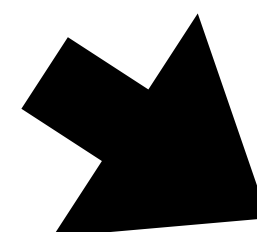
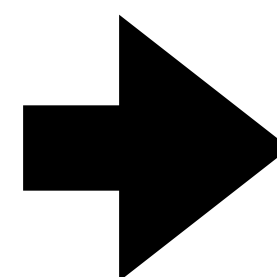
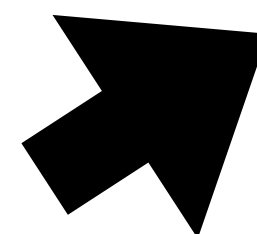
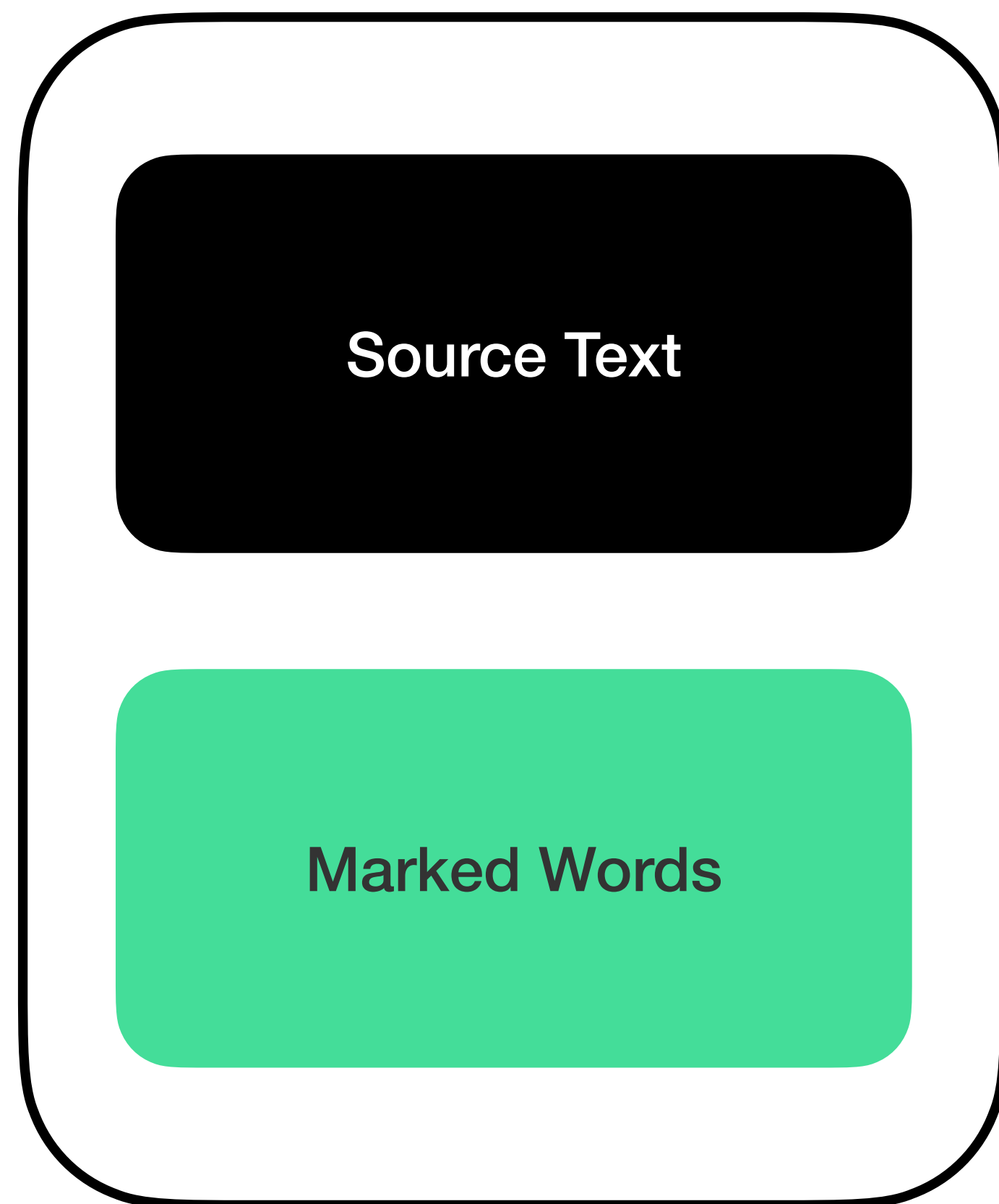


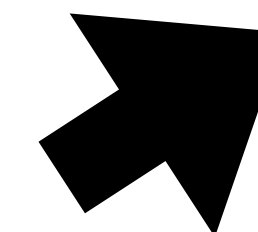
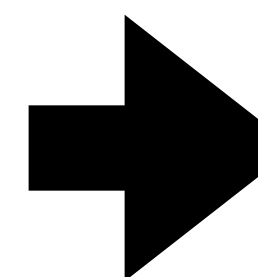
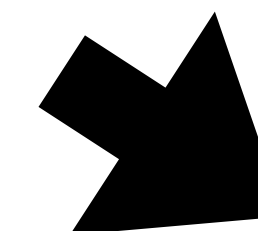
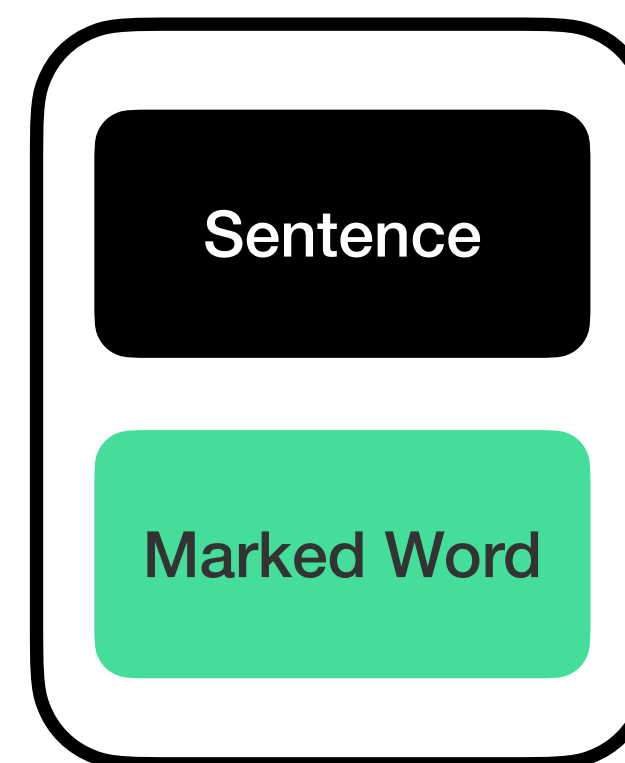
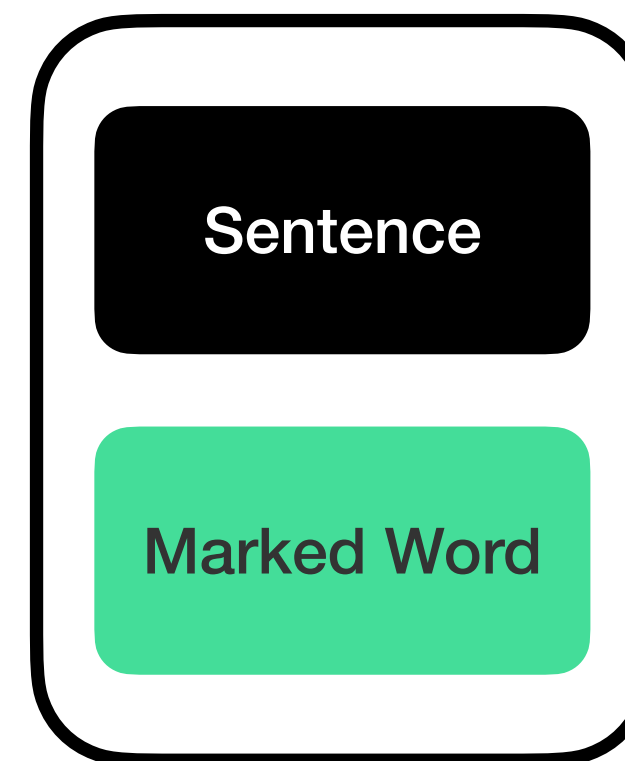
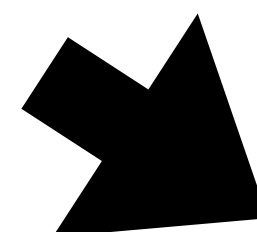
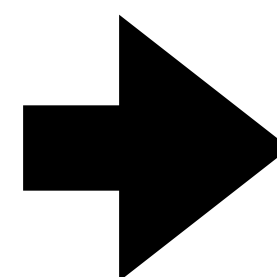
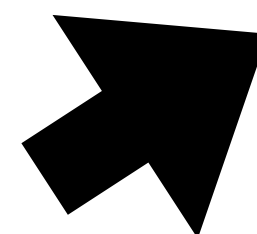
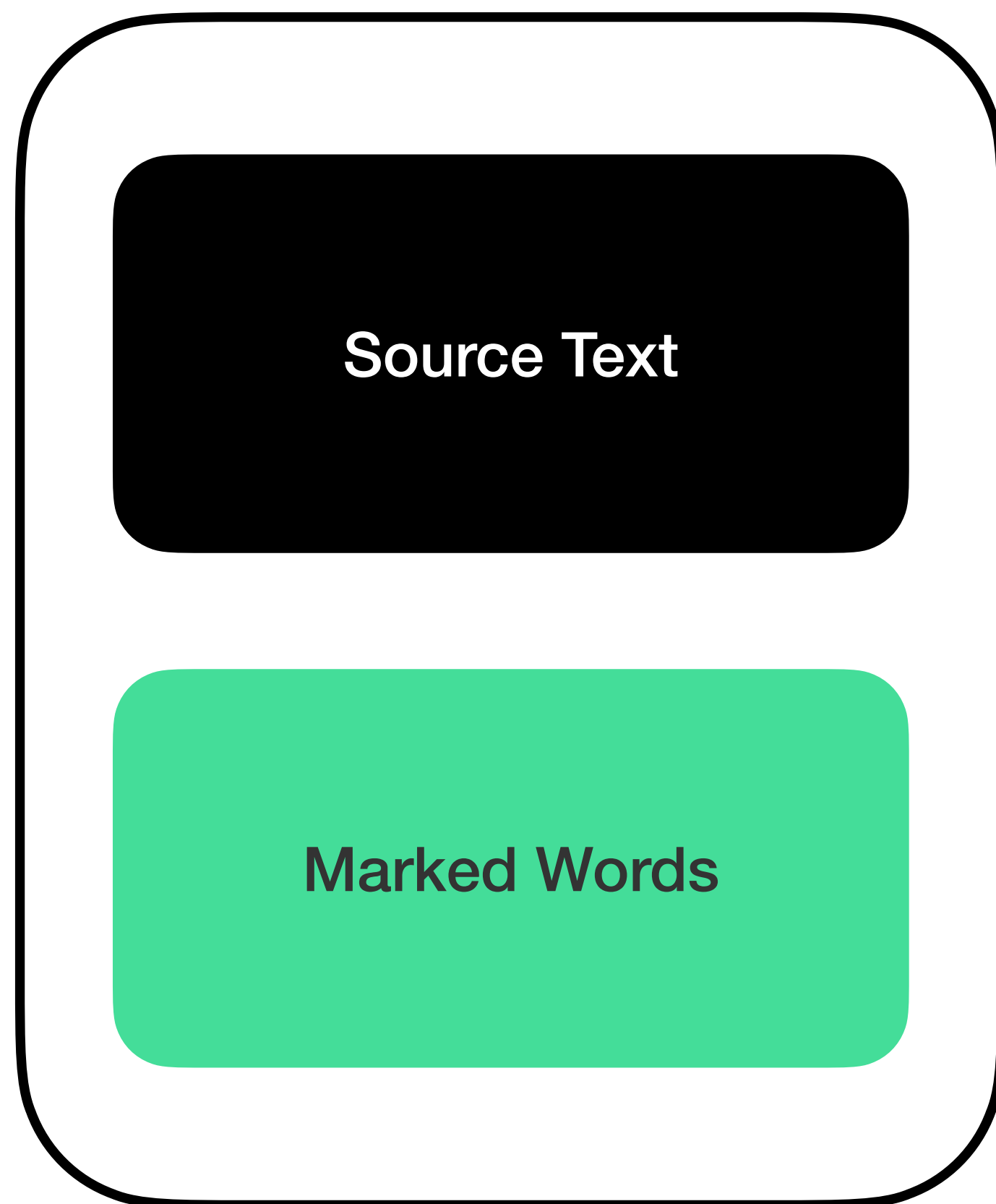
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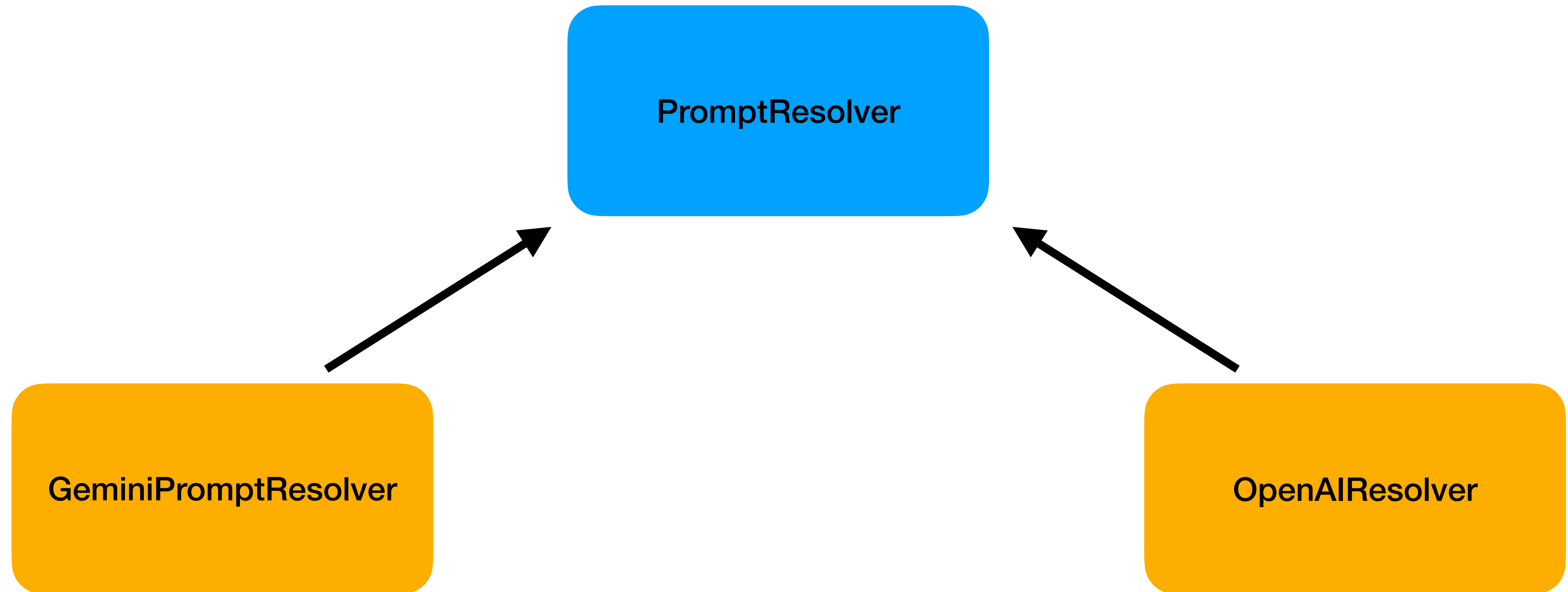




Single vendor LLM
dependency?

Diversify!

```
abstract class PromptResolver {  
    Future<PromptOutput> resolveOutput(  
        List<PromptMessage> messages,  
    );  
}
```



pub.dev

Search Sign in Help

dartantic_ai 1.3.0

Published 40 days ago • [sellsbrothers.com](#)

[INCOMPLETE PACKAGE ANALYSIS]

39

Readme

Changelog

Example

Installing

Versions

Scores

Welcome to Dartantic!

The [dartantic_ai](#) package provides an agent framework designed to make building client and server-side apps in Dart with generative AI easier and more fun!

Key Features

- **Agentic behavior with multi-step tool calling:** Let your AI agents autonomously chain tool calls together to solve multi-step problems without human intervention.
- **Multiple Providers Out of the Box** - OpenAI, OpenAI Responses, Google, Anthropic, Mistral, Cohere, Ollama, and more
- **Streaming Output** - Real-time response generation
- **Typed Outputs and Tool Calling** - Uses Dart types and JSON serialization
- **Multimedia Input** - Process text, images, and files
- **Embeddings** - Vector generation and semantic search

39
LIKES

110
POINTS

561
DOWNLOADS

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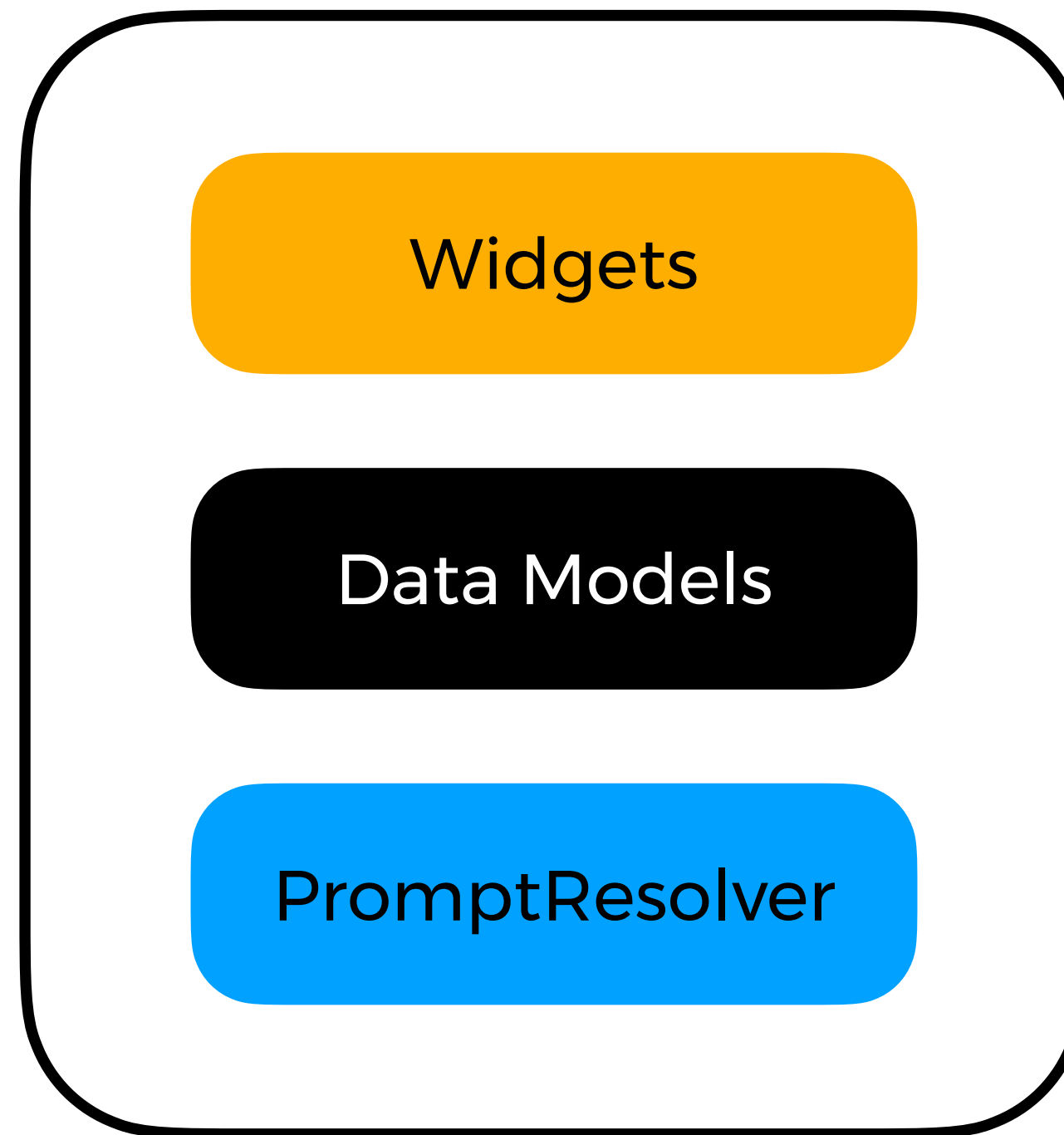
2025.01.12 - 2025.12.07

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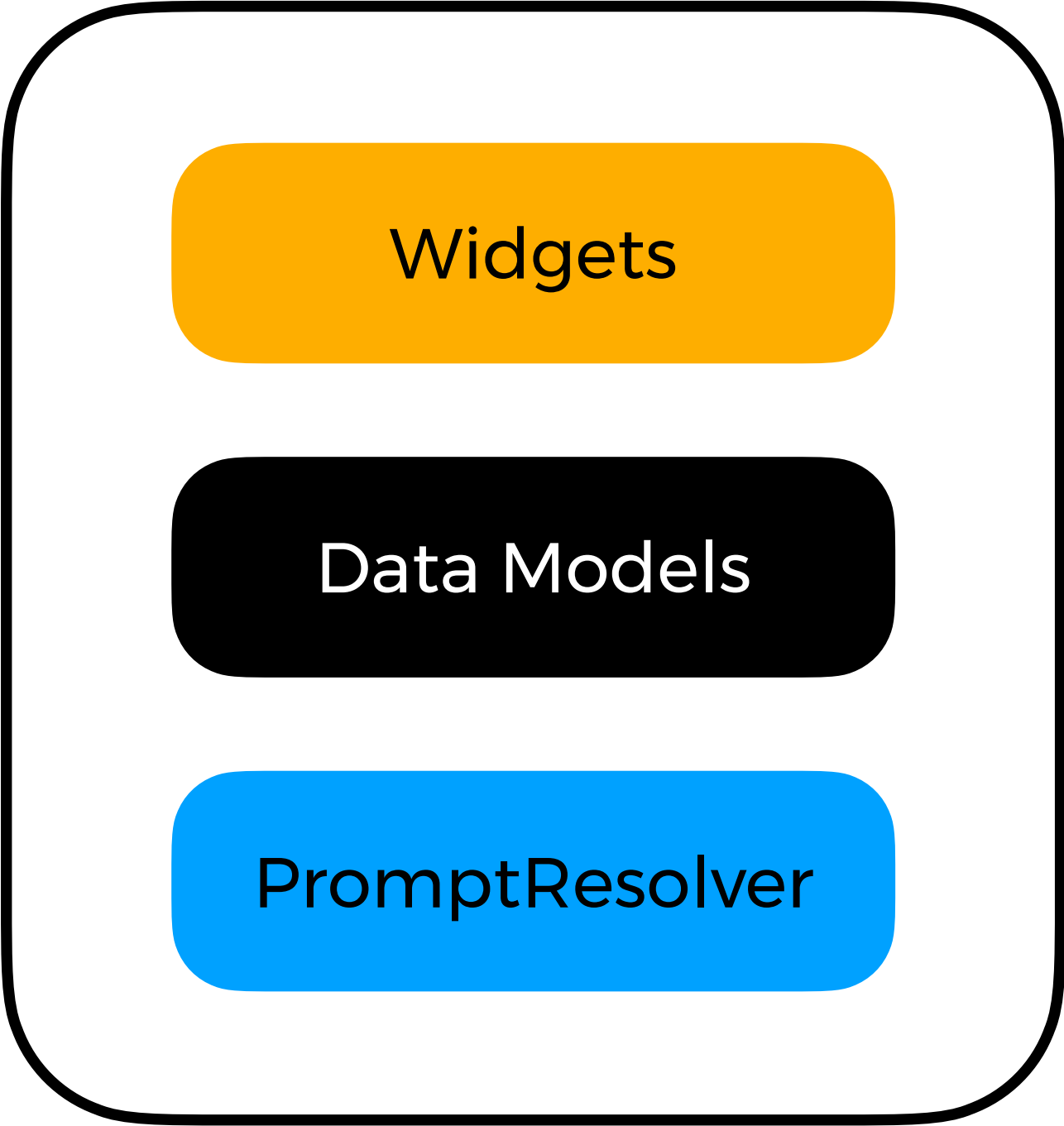
Dartantic is an agentic framework designed to make building client and server-side apps in Dart with generative AI easier and more fun!

Where to put LLM Keys?

Flutter App



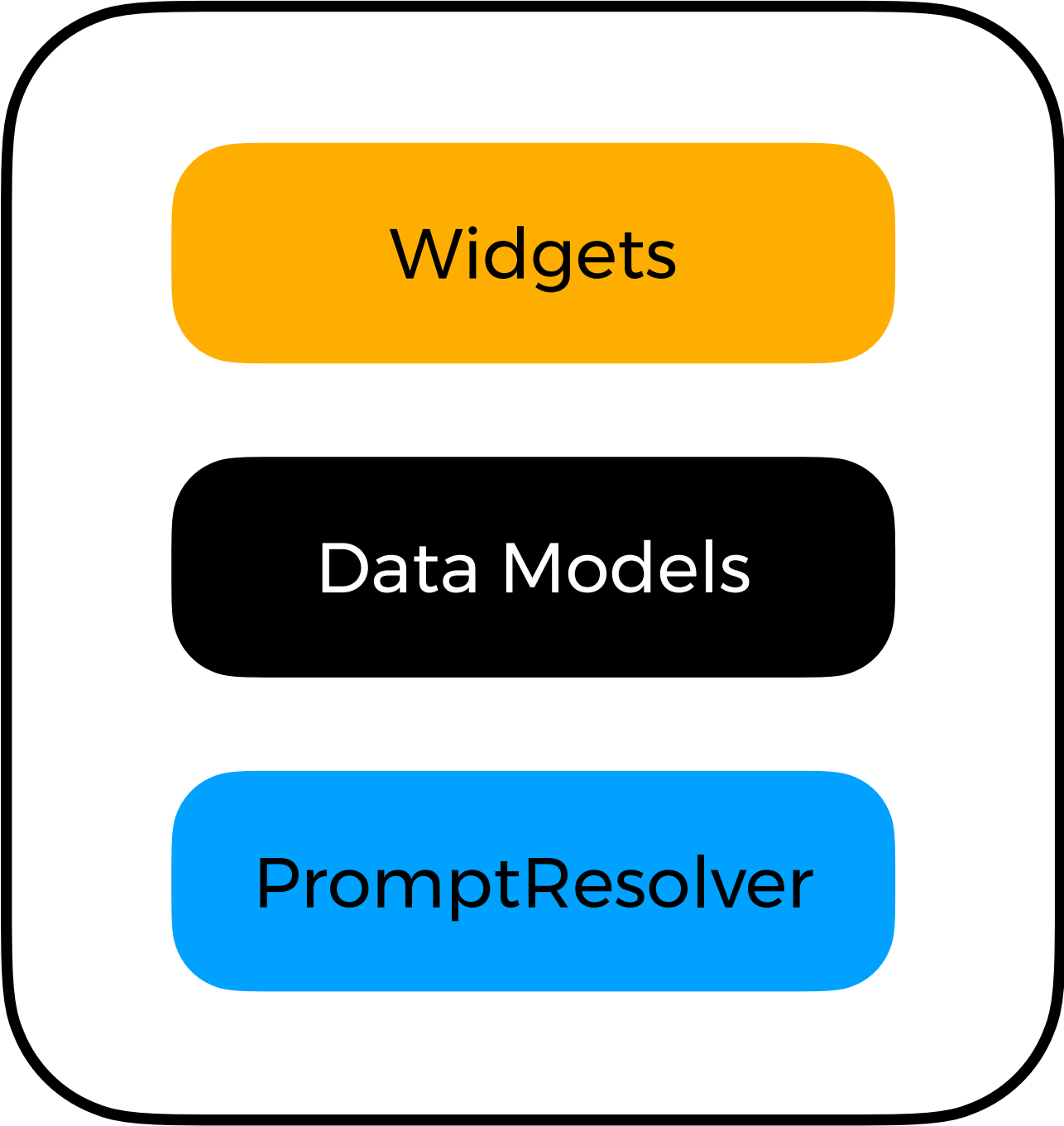
Flutter App



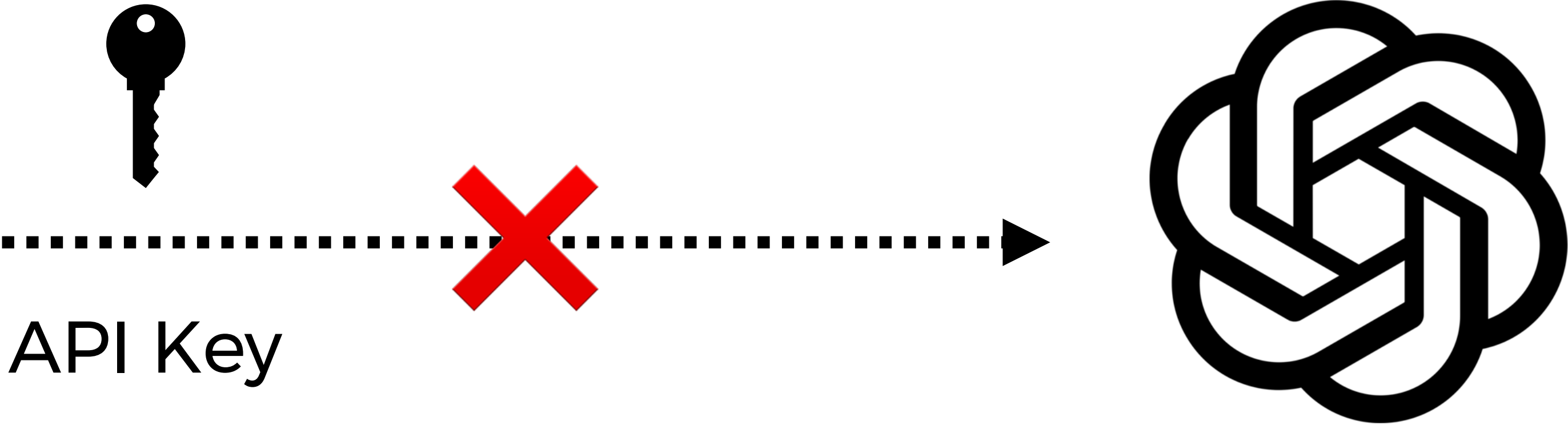
OpenAI



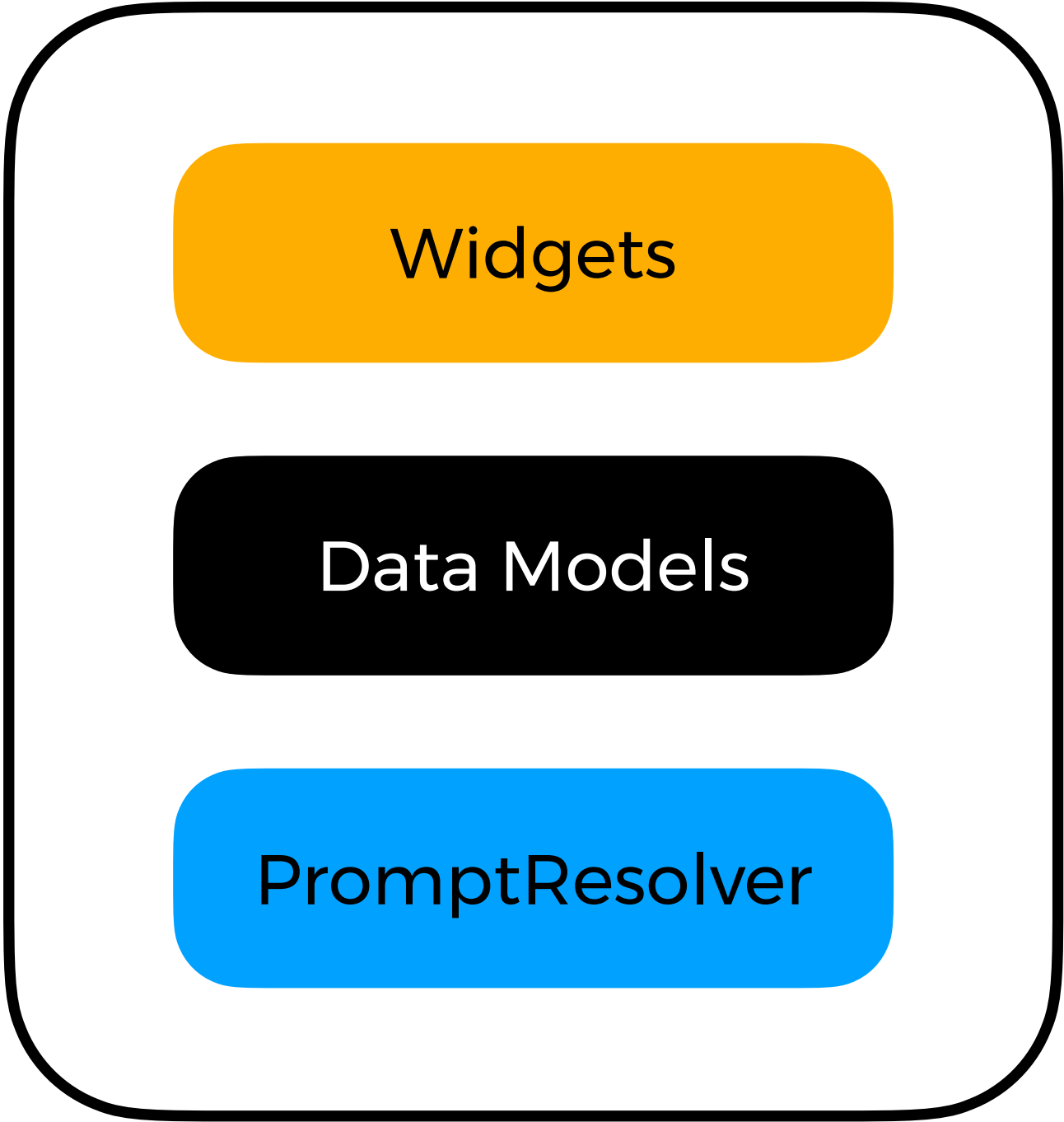
Flutter App



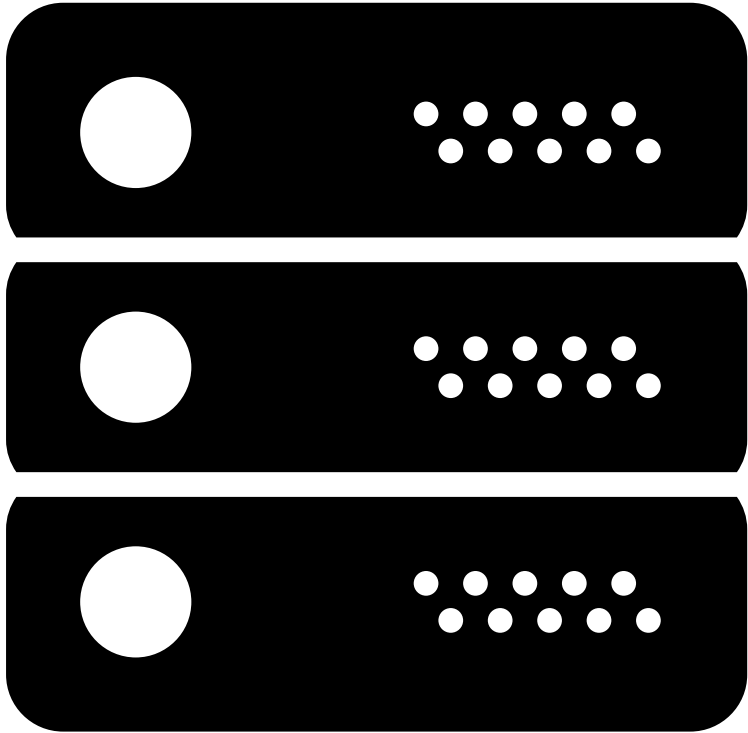
OpenAI



Flutter App



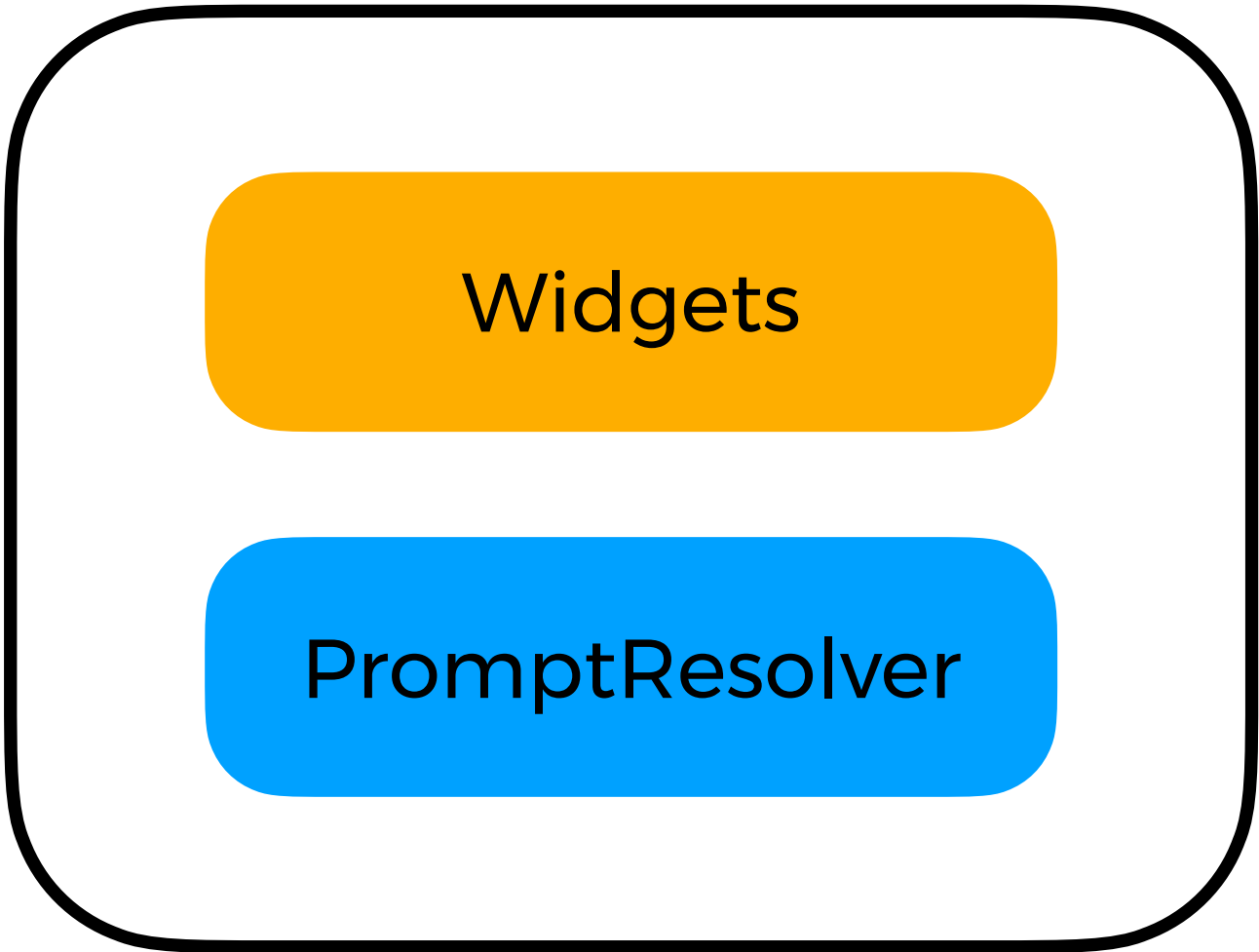
Server



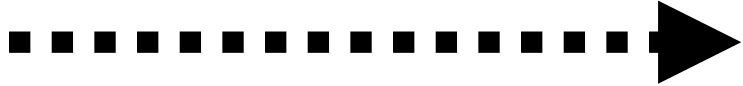
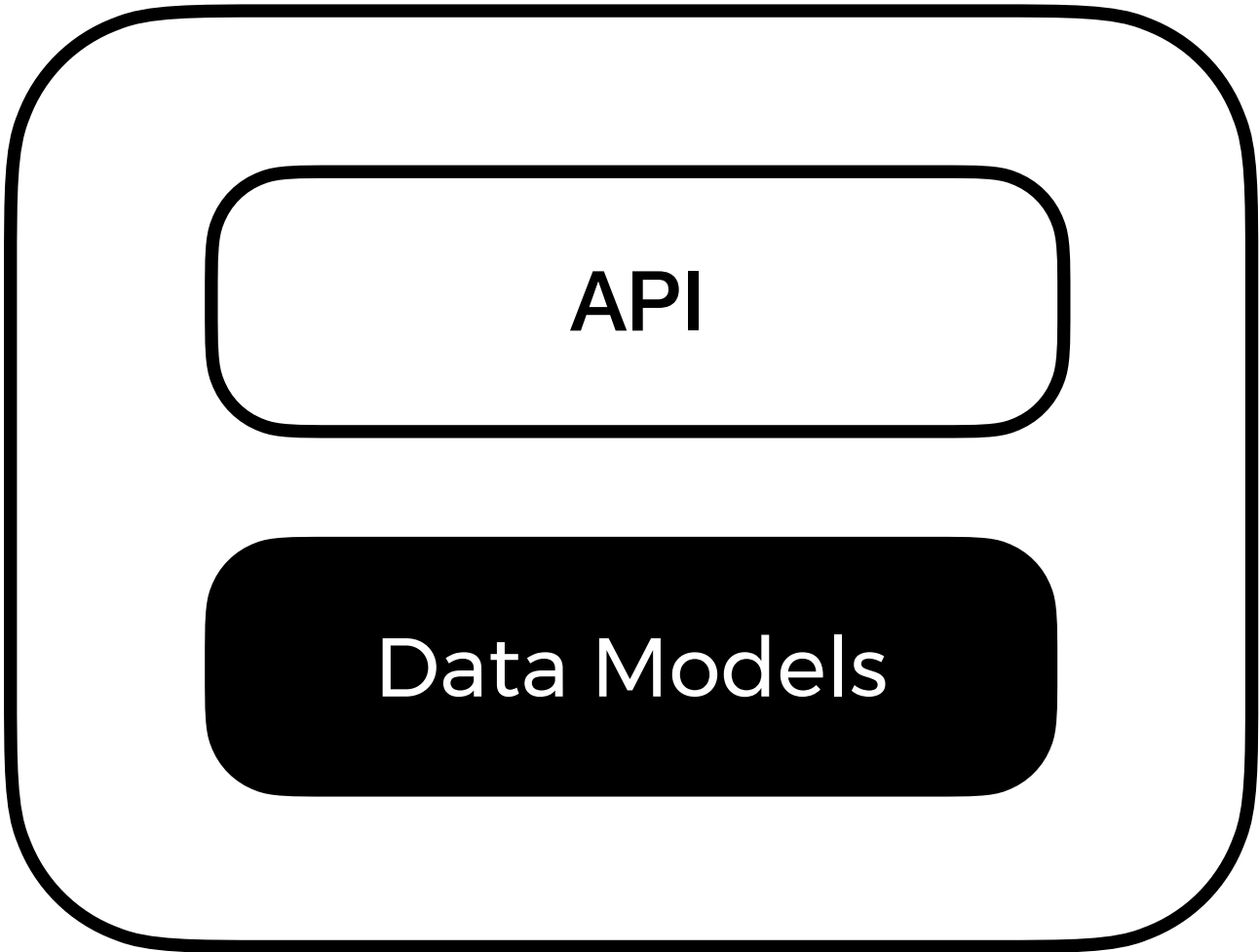
OpenAI



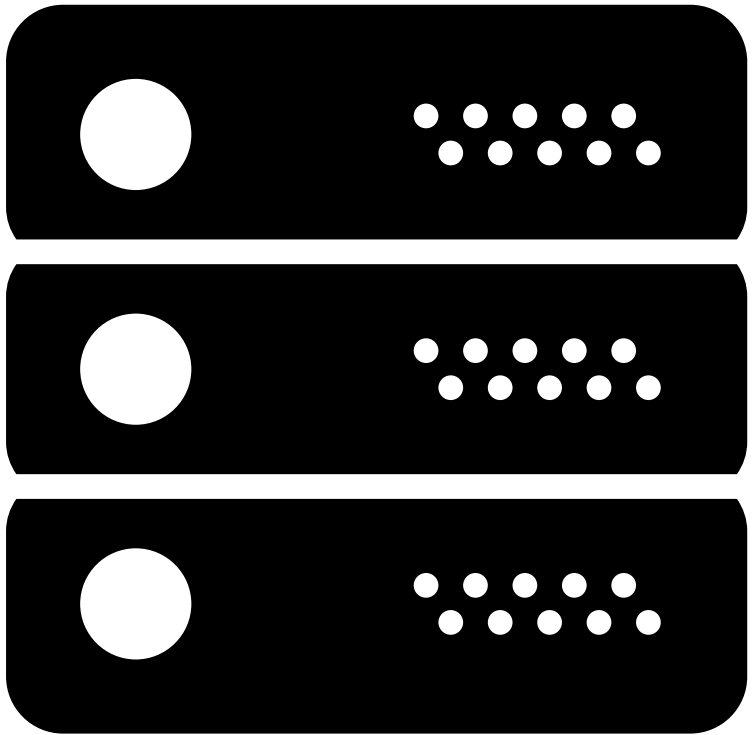
Flutter App



Protocol package



Server

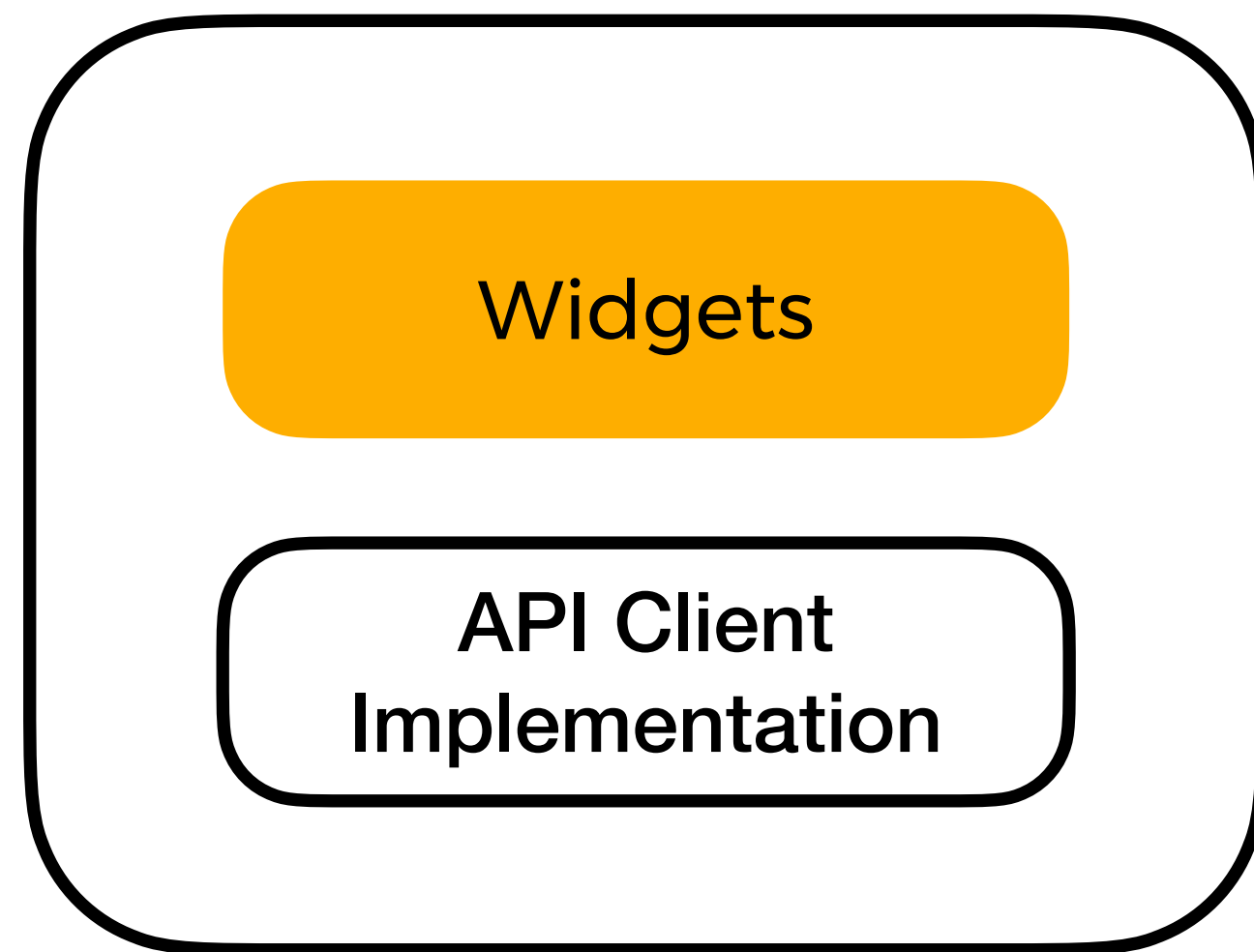


API Key

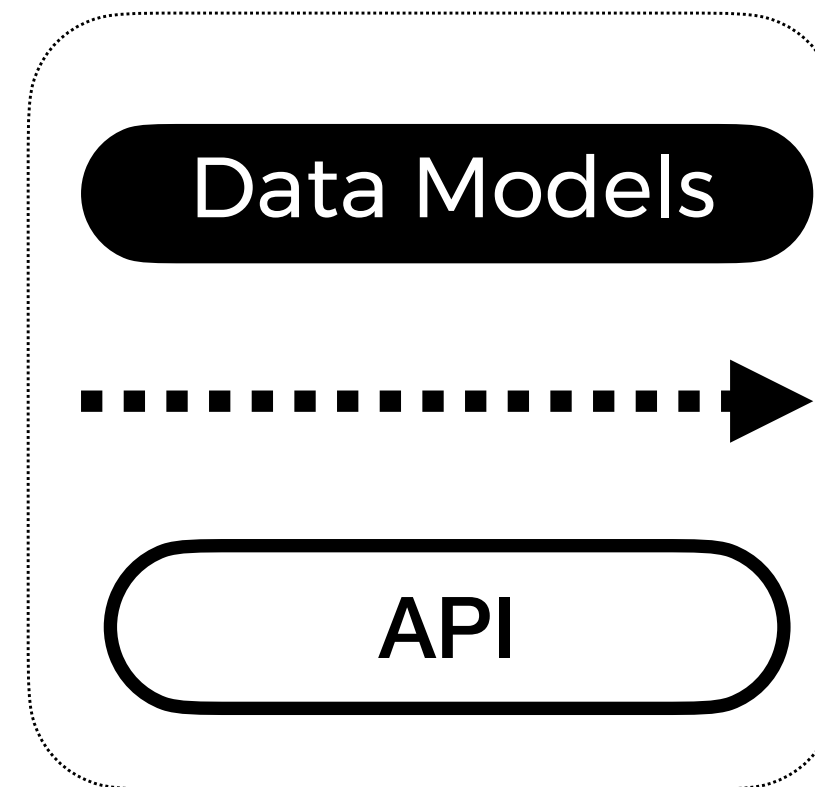
OpenAI



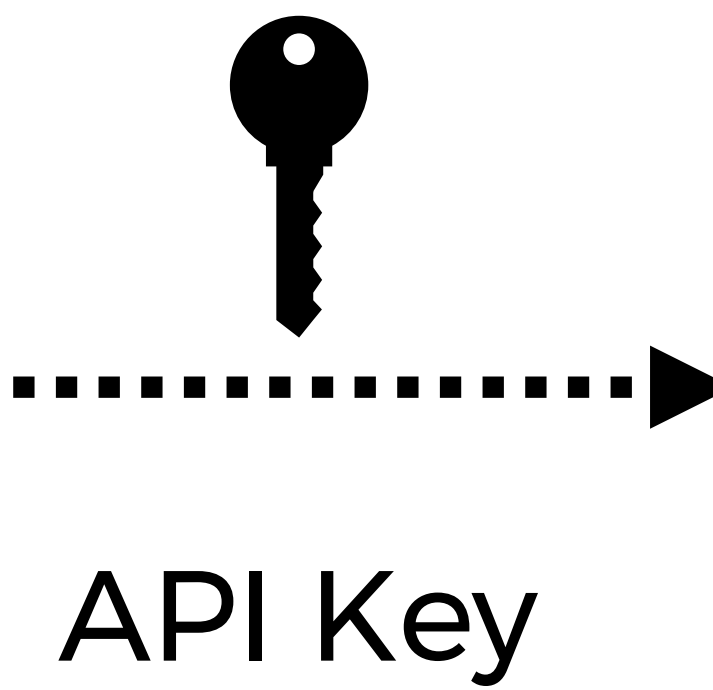
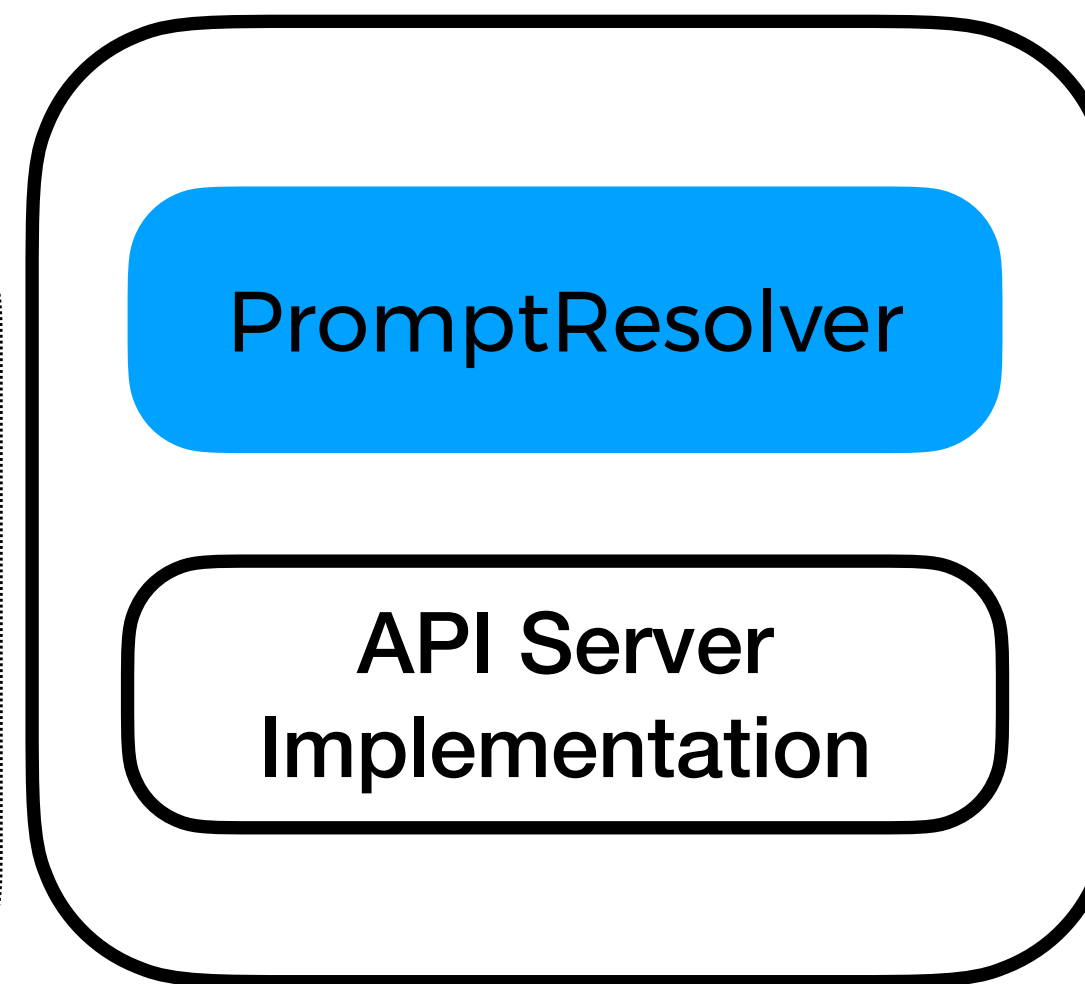
Flutter App



Protocol package



Server



OpenAI



Data Models

Sealed Freezed Data Classes as an API Protocol

```
abstract class API {  
}
```

```
abstract class API {  
    Future<APIResponse> sendRequest(APIRequest request);  
}
```

```
@frozen  
sealed class APIRequest with _$APIRequest {  
}
```

```
@freezed
sealed class APIRequest with _$APIRequest {
  const factory APIRequest.search({
  }) = APIRequestSearch;
}
```

```
@freezed  
sealed class APIRequest with _$APIRequest {  
    const factory APIRequest.search({  
        required String query,  
    }) = APIRequestSearch;  
}
```

```
@freezed
sealed class APIRequest with _$APIRequest {
  const factory APIRequest.search({
    required String query,
  }) = APIRequestSearch;

  factory APIRequest.fromJson(Map<String, Object?> json) =>
    _$APIRequestFromJson(json);
}
```

```
@freezed
sealed class APIRequest with _$APIRequest {
    /// performs search with the given [query]
    const factory APIRequest.search({
        required String query,
    }) = APIRequestSearch;

    factory APIRequest.fromJson(Map<String, Object?> json) =>
        _$APIRequestFromJson(json);
}
```

```
@freezed
sealed class APIResponse with _$APIResponse {
  /// results for [APIRequest.search]
  const factory APIResponse.search({
    required List<String> results,
  }) = APIResponseSearch;

  factory APIResponse.fromJson(Map<String, Object?> json) =>
    _$APIResponseFromJson(json);
}
```

Extension Method(s)

A glue between Request & Response sealed classes

```
abstract class API {  
    Future<APIResponse> sendRequest(APIRequest request);  
}
```

```
extension APIExt on API {  
}
```

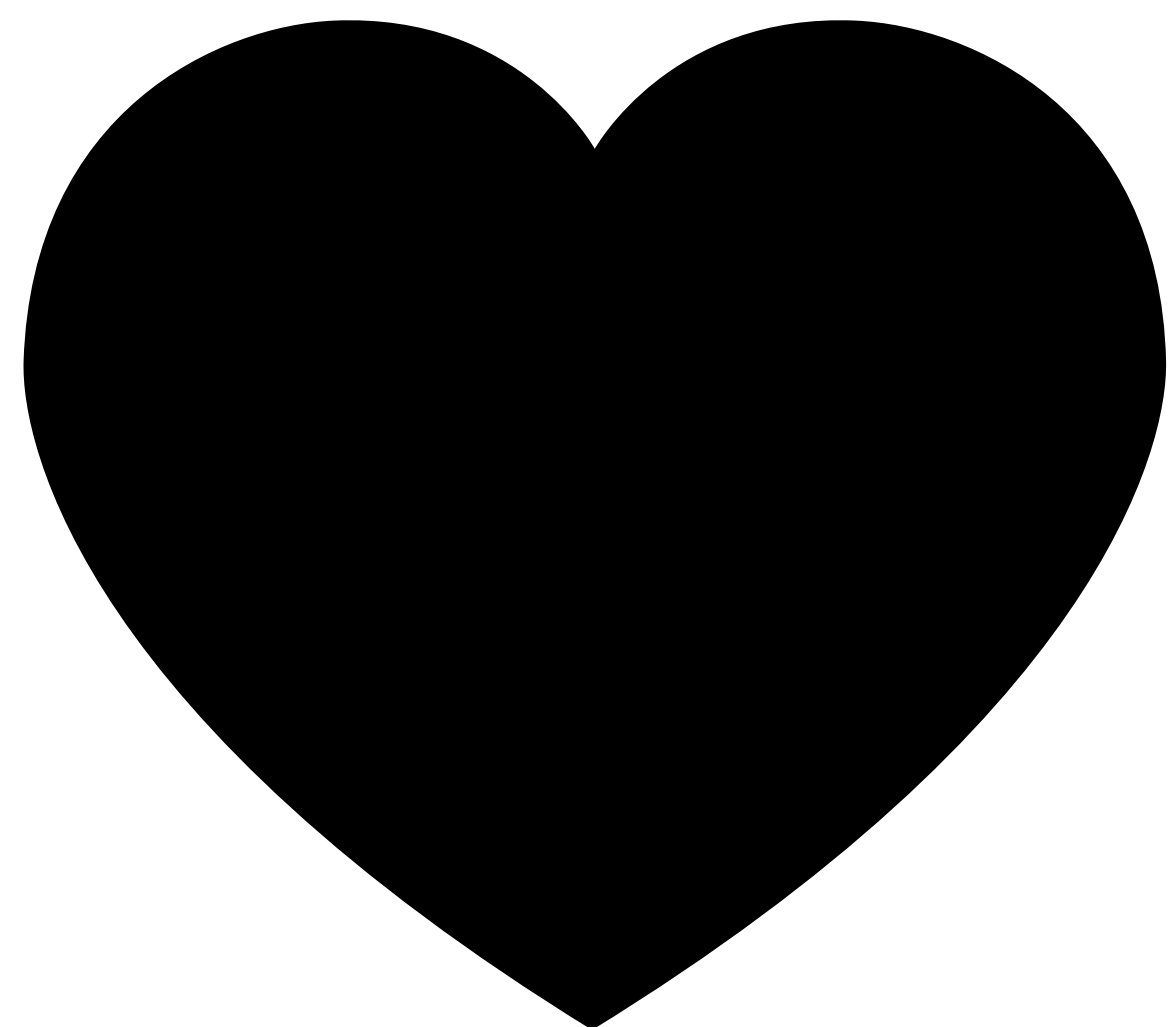
```
extension APIExt on API {  
    Future<List<String>> search(String query) async {  
    }  
}
```

```
extension APIExt on API {  
    Future<List<String>> search(String query) async {  
        final result = await _matches<APIResponseSearch>(  
            APIRequest.search(query: query),  
        );  
  
        return result.items;  
    }  
}
```

```
extension APIExt on API {  
  // ...  
  Future<Res> _matches<Res extends APIResponse>(APIRequest request) async {  
    final result = await sendRequest(request);  
  
    if (result is! Res) {  
      throw StateError("Unexpected type ${result.runtimeType}");  
    }  
  
    return result;  
  }  
}
```

Summary

- Find a project management tool that fits you (FigJam worked for me).
- Get feedback early to confirm your project's usefulness.
- Invest in CI/CD early — it saves time and supports bold changes
- LLMs
 - Break problems into smaller pieces
 - Use multiple AI providers from the start.
 - Prioritize security, even for side projects.



**Thank
You!**



References

[1] Wiredash; <https://wiredash.com/>

[2] Most devs don't understand how context windows work;
<https://www.youtube.com/watch?v=-uW5-TaVXu4>